Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Pricing American-style options requires implied volatility as an input, yet implied volatility is itself derived from observed option prices, a circularity that complicates forward pricing and scenario simulation. We resolved this by separating one-time pooled calibration from forward simulation: the Heston mean-reversion target was reformulated as a function of regime state, days to expiration, moneyness, and an aggregate market-mood indicator, its shape function was calibrated once against a historical option-ladder corpus, and variance was initialized at that mean-reversion target so subsequent forward simulations propagated smile, skew, and term-structure dynamics along each simulated path without consuming additional implied-volatility inputs at simulation time. To recover the daily-fit accuracy of independent per-day surface methods without disturbing the multi-date fit, we added a low-rank daily adapter on the calibrated shape network, scheduled by a streaming median-and-MAD drift trigger; the adapter applied a rank-2 update to the frozen shape surface and adjusted the per-ticker level scalar, leaving forward simulation unchanged. The pricing pipeline combined a Jump Hidden Markov Model that produced regime-aware multi-asset price paths, the modified Heston process that converted each path into an implied-volatility path, and a recombining binomial lattice that priced American options with early exercise. We calibrated the shape function through a hierarchy spanning a five-parameter analytic baseline, a globally shared neural surrogate, and a sector-specific neural surrogate on a multi-ticker, multi-sector option ladder; a temporal-holdout subset isolated scheduled earnings events as the dominant generalization gap, calendar-derived earnings-distance and peer-coupling features recovered the pre-print component, and the low-rank adapter closed the residual day-specific gap across a sweep of high-RMSE cells without perturbing cells where the multi-date fit was already adequate. The framework is released as an open-source Julia package.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices Black and Scholes (1973). This circularity is a practical obstacle for forward pricing and scenario simulation: existing approaches either take IV from the market and feed it into a pricing model at every step of a simulated path, or generate price paths without a consistent IV surface to price contracts against. Parametric IV models such as SABR Hagan et al. (2002) and the SVI parameterization Gatheral (2004) treat the IV surface as an object to be fitted to market quotes, not generated from first principles; they produce smooth interpolations of observed smiles but cannot generate new IV dynamics for unobserved market conditions. Local volatility models Dupire (1994); Derman and Kani (1994) recover a deterministic volatility surface consistent with a single cross-section of option prices, yet the resulting surface is static and cannot produce the stochastic IV fluctuations observed in real markets. Stochastic volatility models such as that of Heston (1993) generate realistic variance paths, but calibrating them requires observed option prices or IV as input, reintroducing the very circularity that a forward pricing pipeline must avoid. The result is a methodological gap: no existing computational pricer separates one-time calibration of an IV-shape function on a historical corpus from forward simulation that propagates a self-consistent IV surface path-conditionally, without ingesting new market-IV inputs along the simulated path.

This gap has direct consequences for forward pricing and scenario simulation. Empirical studies of the IV surface Cont and da Fonseca (2002) have established that smile, skew, and term-structure effects interact with market regimes, so a forward simulator has to reproduce that structure dynamically rather than freeze a single cross-section. Calibrating Heston or SABR to a market snapshot Hagan et al. (2002); Gatheral (2004) produces a self-consistent surface at one instant but locks the parameters to that snapshot, so forward-simulated option prices drift from the structural return dynamics whenever the underlying regime moves. A multi-date calibration of a shared shape function on a historical ladder corpus, by contrast, fixes the calibration cost up front and lets the forward simulator propagate IV without re-ingesting market quotes at each step. Historical resampling Glasserman (2004) preserves realized distributional features but cannot produce out-of-sample tail scenarios consistent with a structural model of returns, and generative neural approaches Wiese et al. (2020) reproduce equity stylized facts without producing self-consistent option-price and IV pairs, so any downstream pricing step has to reintroduce the very calibration loop the simulator was meant to avoid. A practitioner who wants path-conditional American option prices, finite-difference Greeks, and forward IV from one coherent simulation therefore has to either accept the static-surface compromise or stitch together inconsistent models Buehler et al. (2019), neither of which serves scenario analysis or backtesting pipelines that depend on option-level outputs.

In this study, we developed a computational pricing pipeline in which IV emerged as a path-conditional *output* of forward simulation, after the IV-shape function was calibrated once on a historical option-ladder corpus and held fixed thereafter. The pipeline coupled a Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner (2026) that produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont (2001), with cross-asset dependence imposed via a Student-t copula Demarta and McNeil (2005); a modified Heston stochastic variance process Heston (1993) that converted those price paths into

IV paths, where the mean-reversion target θ was a function of the HMM regime state, days to expiration (DTE), moneyness, and an aggregate market-mood indicator; and a Cox-Ross-Rubinstein (CRR) binomial tree Cox et al. (1979) that converted the pre-computed IV into American option prices with early exercise. The central innovation was a hybrid θ-function that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics. An equilibrium initialization $v_0 = \theta(t = 0)$ gave each (ticker, strike, DTE) triple its own initial IV; once the shape function was calibrated against the historical ladder corpus, the forward simulator propagated smile, skew, and term-structure dynamics path-conditionally without re-ingesting market IV along the simulated path. Within this pipeline, the shape function that controlled the smile and term structure of θ admitted a family of representations, and the modeling choice at each level of the family traded off model capacity for sharing across tickers. We therefore calibrated the shape function through a hierarchy of increasingly expressive forms: a parsimonious five-parameter functional form on single-ticker and cross-ticker semiconductor option chains, a globally shared neural surrogate that replaced the analytic form with a data-driven representation, and a sector-specific neural surrogate that let within-sector shapes specialize. The same decomposition supported a further relaxation to per-ticker networks once per-name sample sizes grew, without changing the downstream pipeline. We demonstrated this progression on a 31-ticker, fifteen-date, six-sector option ladder and on an eight-day temporal-holdout subset of the same capture, the latter of which isolated scheduled earnings events as the dominant source of the temporal generalization gap and motivated calendar-derived earnings-distance and same-sector peer-distance features that recovered a portion of that gap. The framework also produced realistic forward IV paths and American option prices with an endogenous crash-protection premium. The pipeline was implemented as an open-source Julia package so that the calibration, forward simulation, and pricing steps could be reproduced and reused end to end.

2 Background

The framework built on four established components, each contributing a distinct piece of the synthetic option generation pipeline: the JumpHMM produced realistic multi-asset price paths; the Heston stochastic variance process provided a mechanism for converting price dynamics into implied volatility; a recombining binomial lattice translated IV into American option prices with early exercise; and a neural surrogate replaced the analytic shape function for the IV surface with a data-driven representation that absorbed sector-specific smile and term-structure geometry.

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner (2026) was developed for generating synthetic equity time series that simultaneously preserved three stylized facts of financial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering Cont (2001). The model discretized continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. The hidden state $S_t \in \{1, \ldots, N\}$ at each timestep governed the distribution of the excess growth rate G_t; each state k emitted observations from a location-scale Student-t distribution: $G_t \mid S_t = k \sim \mu_k + \sigma_k \cdot t_\nu$, where μ_k and σ_k were learned per-state location and scale parameters and ν was the shared degrees-of-freedom parameter. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes that produced the volatility clustering characteristic of empirical return series. For multi-asset generation, cross-asset dependence was imposed via a Student-t copula Demarta and McNeil (2005) that preserved each asset's marginal distribution while introducing realistic tail dependence; this copula structure ensured that joint tail events, in which multiple assets simultaneously occupied extreme states, occurred more frequently than a Gaussian copula would imply.

2.2 Heston stochastic variance model

The Heston model Heston (1993) specified the instantaneous return variance v of the underlying asset (i.e., the variance rate of its log-returns) as a mean-reverting square-root process:

$$dv = \kappa(\theta - v)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{1}$$

where κ was the mean-reversion speed controlling how quickly the variance returned to its long-run level, θ was the long-run variance target, and σ_v was the vol-of-vol governing the amplitude of stochastic fluctuations in v. In the standard formulation, the asset price and variance processes were coupled through a correlation ρ between their driving Brownian motions, producing the leverage effect in which negative returns coincided with rising volatility. The key property for our purposes was mean-reversion: the variance process was pulled toward θ, so by controlling θ we could control the equilibrium level of implied volatility. More generally, the Heston model produced a full European IV surface in (K, τ) through its semi-closed-form characteristic-function pricing, and this analytic surface was the shape function that the neural surrogate below replaced in our pipeline. In the standard model θ was a constant; our extension made θ time-varying and state-dependent, linking it to the JumpHMM regime state, contract characteristics, and market conditions.

2.3 Lattice models for American options

Once the Heston shape function above supplies an IV per contract, the pipeline still needs to convert $(S, K, \tau, r, q, \sigma)$ into an option price; for American options this rules out closed-form Black-Scholes and requires a numerical scheme. Lattice methods discretize the underlying-price process onto a recombining tree and price contingent claims by backward induction with an early-exercise check at each node Broadie and Detemple (1996). The Cox-Ross-Rubinstein (CRR) tree Cox et al. (1979) is the canonical construction, with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. Subsequent variants modified the moment-matching to address specific shortcomings of the CRR construction: Jarrow and Rudd (1982) matched the first two moments under the risk-neutral measure with $p = 1/2$ at the cost of a non-zero log-drift term, the modified lattice of Tian (1993) added a third-moment match to suppress the order-$\sqrt{\Delta t}$ pricing bias of the CRR tree, and the Leisen-Reimer (LR) tree of Leisen and Reimer (1996) replaced the moment-matching construction altogether with a Peizer-Pratt inversion of the binomial CDF, choosing p and u at each step so that the strike K landed on a tree node by design at every (S, σ) configuration. All four constructions converge to the Black-Scholes price as the number of steps increases, but their finite-step behavior differs sharply: the CRR price oscillates around the continuum limit because K drifts across node boundaries as Δt shrinks, the LR price converges monotonically because the strike is pinned to a node, and the Tian and Jarrow-Rudd trees fall in between Leisen and Reimer (1996); Broadie and Detemple (1996). The pinned-strike property of the LR construction also stabilizes finite-difference Greeks: small bumps to S or σ leave the strike on a node, so the staircase aliasing that CRR exhibits when the bump moves K across a node boundary is absent.

Two of these lattices appear later in this work and the choice between them is dictated by the use case. The pricing-pipeline pricer is the CRR tree at 200 steps: with IV pre-computed by the upstream Heston shape function, the pricer needs only a single σ input per contract, and CRR's recombining structure makes the cost of pricing the full 234,549-row ladder linear in the number of steps. The forward-scenarios pricer, in contrast, required smooth Greeks at modest tree depth because the per-path scenarios in the short-premium analysis computed Δ, Γ, and Vega by central finite differences against thousands of (S_t, σ_t) configurations along each simulated path; we therefore switched to the LR tree at 201 steps for that purpose, where the pinned-strike construction eliminated the bump-induced aliasing that CRR produced at the same depth. The pipeline did not depend on either choice in particular: any of the four lattices above would have been a drop-in replacement for the appropriate stage, and the decomposition between IV calibration and option pricing kept the choice of tree orthogonal to the rest of the framework.

2.4 Neural representations of the implied volatility surface

Parametric IV models such as SABR Hagan et al. (2002) and SVI Gatheral (2004) expressed the smile and term structure through low-dimensional closed-form functions, chosen to interpolate observed quotes while respecting no-arbitrage constraints. These forms were parsimonious and interpretable but inherited their expressive capacity from the chosen functional family, and a single global parameterization across a broad universe of tickers could not simultaneously accommodate sector-specific skew patterns and smile curvatures. A complementary line of work Horvath et al. (2021); Bayer and Stemper (2018) replaced the analytic form with a neural network that took $(\ln \tau, \ln m)$ or a similar low-dimensional set of contract and model features as input and returned an IV or variance output; the network provided a flexible universal approximator for the shape of the surface while the downstream pricing pipeline remained unchanged. Neural surrogates of this kind have been used both as fast pricing approximators and as calibration targets in deep calibration schemes, and they admit the same multiplicative decomposition between a per-ticker level and a shared shape that the parametric forms use, making them a drop-in replacement for the functional form inside a generative pipeline. We used the neural surrogate in the structural sense rather than the deep-calibration sense: ψ_{NN} was the shape factor inside the multiplicative decomposition whose level v_t evolved with a JumpHMM-coupled Heston process, so the network supplied the cross-sectional geometry while the dynamics came from the generative model.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\begin{aligned}&\underbrace{\text{JumpHMM(prices)}}_{\text{Stage 1}} \;\rightarrow\; \{S_t,\, s_t,\, \text{jumps}\} \;\rightarrow\; \underbrace{dv \text{ process}}_{\text{Stage 2}} \;\rightarrow\; \sigma^2_{\text{imp}} = v_t \\ &\rightarrow\; \underbrace{\text{CRR}(S_t,\, \sigma_{\text{imp}},\, K,\, \text{DTE},\, r_f)}_{\text{Stage 3}} \;\rightarrow\; P_{\text{american}}\end{aligned} \tag{2}$$

This ordering was dictated by data flow: the JumpHMM produced the price paths and regime-state sequences that the Heston process required as inputs, and the Heston process produced the IV values that the CRR tree required for pricing. Each stage was self-contained, so the pipeline could be run forward without iteration or feedback from downstream stages.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa\big(\theta(t) - v\big)\,dt + \sigma_v\sqrt{v}\,dW_v \tag{3}$$

The mean-reversion target decomposed into a per-ticker base level, an aggregate market mood modulator, and a contract-specific smile and term-structure adjustment that included a scheduled-event indicator:

$$\theta(i,t) = \theta_{i,\,s_t} \cdot \big(1 + \gamma \cdot M_t\big) \cdot \psi\big(\text{DTE}_t,\; K/S_t,\; e_{i,t},\; e^{\text{peer}}_{i,t}\big) \tag{4}$$

where $\theta_{i,\,s_t}$ was a per-ticker, regime-dependent variance level (one value per ticker i per HMM state s_t); $\gamma \geq 0$ was the market mood sensitivity controlling how strongly aggregate stress elevated IV; $M_t \in [0,1]$ was the aggregate market mood; and ψ was the smile, term-structure, and event-shape adjustment with two earnings-event inputs $e_{i,t}$ (signed days from t to ticker i's nearest earnings print) and $e^{\text{peer}}_{i,t}$ (the minimum of $|e_{j,t}|$ over same-sector equity peers $j \neq i$). The regime-dependent base $\theta_{i,\,s_t}$ linked the Heston process directly to the JumpHMM state sequence for forward simulation: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), $\theta_{i,\,s_t}$ rose, elevating the IV target. This mechanism produced the leverage effect as an emergent property of the return dynamics rather than as a separate model component: large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ. For the

IV-surface calibration that follows, we restricted (4) to the operational form

$$\theta_{\mathrm{cal}}(i,t) = \theta_i \cdot \psi\big(\mathrm{DTE}_t,\ K/S_t,\ e_{i,t},\ e_{i,t}^{\mathrm{peer}}\big) \tag{5}$$

in which $\gamma = 0$ (the aggregate mood multiplier was reserved for forward simulation with simulated state paths) and the per-state level $\theta_{i,\,s_t}$ collapsed to a single per-ticker level θ_i (since the calibration corpus consisted of observation-day IV surfaces rather than state-conditioned histories). Both restrictions were transparent to ψ, so the shape function carried over without modification when the full state-and-mood mechanism was reactivated for simulation.

The function ψ captured the shape of the IV surface with five parameters:

$$\psi(\tau,\ m) = \exp\Big(\beta_1 \ln\tau + \beta_2 \ln m + \beta_3 \ln\tau\ \ln m + \beta_4(\ln m)^2 + \beta_5(\ln\tau)^2\Big) \tag{6}$$

where $\tau = \max(\mathrm{DTE}, 1)$ and $m = K/S$. The parameter β_1 controlled the linear term-structure decay; β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE-skew interaction (the skew flattened at longer maturities); β_4 controlled the smile curvature; and β_5 controlled the DTE curvature, capturing the U-shaped ATM term structure where short-dated and long-dated IV were both elevated relative to intermediate maturities.

Rather than specifying a separate initial variance v_0, we initialized the variance process directly at the per-contract mean-reversion target so that the entry IV surface inherited the smile, skew, and term structure encoded in ψ without a separate IV-surface input at simulation time:

$$v_0 \ = \ \theta(i,\,0;\,K) \ = \ \theta_{i,\,s_0}\,\big(1+\gamma M_0\big)\,\psi\big(\mathrm{DTE}_0,\ K/S_0,\ e_{i,0},\ e_{i,0}^{\mathrm{peer}}\big) \tag{7}$$

$$\sigma_t^2(K,\,S_t) \ = \ v_t \tag{8}$$

The per-contract IV was the variance state itself: ψ entered the model only through the time-varying mean-reversion target $\theta(i,t;K)$ in (4) that drove the SDE drift, not as a separate multiplicative factor on v_t, so v_t was a per-contract process and every (ticker, K, DTE) triple obtained its own variance trajectory under the same Brownian forcing with a contract-specific drift target. By construction $\sigma_0^2 = v_0 = \theta(i,0;K)$ recovered the calibrated decomposition (11) at the contract's entry capture, and the variance process immediately began mean-reverting around the same $\theta(i,t;K)$ at every subsequent step as the regime state, aggregate mood, and moneyness evolved. The value of this design choice lay in eliminating v_0 as a free parameter: the entry-time IV was determined by construction from θ_i and the calibrated ψ surface, with ψ itself trained once on the multi-date ladder corpus via the loss in (13).

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states, where N was the JumpHMM state count and N_{tail} denoted the number of extreme states at each end of the state space (inherited from the JumpHMM fit):

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\Big\{ s_t^{(i)} \le N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}} \Big\} \tag{9}$$

This signal was fully endogenous to the JumpHMM copula model; it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts, so that correlated IV spikes during market stress events emerged by construction. The variance process was discretized via Euler-Maruyama with a reflecting boundary at zero and a daily time step $\Delta t = 1/252$:

$$v_{t+1} = \Big| v_t + \kappa\big(\theta_t - v_t\big)\Delta t + \sigma_v \sqrt{\max(v_t, 0)}\,\sqrt{\Delta t}\, Z_t \Big|, \quad Z_t \sim \mathcal{N}(0,1) \tag{10}$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen (2008), a practical necessity since tail states could drive θ to values where the Feller condition would otherwise have been violated. American option prices were computed from the resulting IV using a CRR binomial tree with 200 steps per contract, a depth chosen because the price oscillation amplitude across that range fell below \$0.05 on the contracts of interest, well inside the bid-ask spread of the underlying market.

3.3 Neural surrogate for ψ

The parametric form of ψ in equation (6) used five shape parameters shared across all contracts on a given ticker. When the model was applied to a broad universe spanning multiple sectors, this global shape became a bottleneck: the semiconductor smile geometry differed systematically from mega-cap ETFs, healthcare names carried idiosyncratic event risk, and energy tickers inherited commodity-driven skew patterns not expressible by a five-term log-polynomial. We therefore replaced the parametric ψ with a neural surrogate Horvath et al. (2021), keeping the multiplicative decomposition

$$\sigma^2_{\text{model}} \;=\; \theta_{\text{ticker}} \cdot \psi \tag{11}$$

in which σ_{model} was the model's per-contract implied-volatility prediction at strike K and time-to-maturity τ on capture date t, expressed in the same Black-Scholes convention the market data providers used to report IV on the option chain, fitted against $\text{IV}_{\text{market}}$ by the loss in (13), and supplied to the lattice pricer as its single diffusion-volatility input, so σ_{model} played the dual role of calibration target and pricing-stage diffusion vol. We allowed the shape to be data-driven and added two scheduled-event inputs (e, e^{peer}) that captured pre-earnings IV expansion and same-sector earnings spillover:

$$\psi_{\text{NN}}(\ln\tau,\, \ln m,\, e,\, e^{\text{peer}}) = \exp\!\big(\text{NN}(\ln\tau,\, \ln m,\, e,\, e^{\text{peer}})\big) \tag{12}$$

where each input was standardized to its training-set mean and standard deviation before entering the network. The earnings inputs were the signed clipped days-to-earnings for the contract's ticker, $e = \text{clip}(\Delta^{(i)}_{\text{earn}}, -30, 30)$, and the minimum absolute days-to-earnings over same-sector equity peers excluding self, $e^{\text{peer}} = \min_{j \neq i,\, c(j)=c(i)} |\Delta^{(j)}_{\text{earn}}|$ (clipped to 30). The peer feature carried the sector-coupling signal: a ticker j in the same sector printing on day t contributed $e^{\text{peer}} = 0$ for all of i's contracts that day, which let the network learn the IV-spillover shape characteristic of the sector. For ETFs (no own earnings), e was set equal to e^{peer} taken over the full equity universe. The network produced $\ln \psi$ so that $\psi > 0$ by construction, preserving the log-Gaussian interpretation of the smile surface used in the parametric form. Each sector network was a two-hidden-layer multilayer perceptron with tanh activations: $4 \to 16 \to 16 \to 1$ (369 parameters) for groups with ≥ 2000 observations and $4 \to 8 \to 8 \to 1$ (121 parameters) for smaller groups, a width chosen to keep the observations-to-parameters ratio above 30 across all sector fits.

The decomposition defined a natural hierarchy of sharing. At one extreme, a single global ψ_{NN} shared across all tickers assumed that smile geometry was universal and left only the variance level θ_{ticker} to distinguish names. At the other extreme, a per-ticker ψ_{NN} recovered ticker-specific shape but required enough data per name to resolve the surface without overfitting. We adopted a sector-specific intermediate: one $\psi^{(c)}_{\text{NN}}$ per sector c, with all tickers within a sector sharing shape but each ticker retaining its own θ_{ticker}. This choice reflected the empirical observation that within-sector tickers shared dominant risk factors (technology names tracked semiconductor cycles, healthcare names tracked FDA and trial catalysts, financials tracked rate expectations), so pooling shape across a sector provided a favorable bias-variance tradeoff at the current data volume. As per-ticker sample sizes grew, the same architecture supported relaxation to per-ticker networks without changing the downstream pipeline.

Training minimized the mean squared IV error jointly over the network weights and the per-ticker log-variance levels $\{\ln \theta_{\text{ticker}}\}$:

$$\min_{w,\, \{\ln \theta_t\}} \frac{1}{n} \sum_{i=1}^{n} \left(\sqrt{\theta_{t(i)} \cdot \psi_{\text{NN}}(\ln \tau_i,\, \ln m_i,\, e_i,\, e_i^{\text{peer}};\, w)} - \text{IV}_{\text{market},i} \right)^2 \tag{13}$$

where $t(i)$ denoted the ticker of the ith observation. The log-variance levels were initialized from each ticker's mean squared market IV and then updated jointly with the network weights under a single Adam optimizer, so the level and shape fits converged together rather than alternating fits of one conditional on the other. The learning rate followed a step schedule (1e-3 $\to$ 5e-4 at epoch 500 $\to$ 2e-4 at 1000 $\to$ 1e-4 at 1500), and training ran up to 2000 epochs with a best-checkpoint patience of 200 epochs on the training loss. Earnings dates were sourced from the public Yahoo Finance calendar and joined to each contract by ticker and observation date.

3.4 Daily low-rank adapter

Each Dense layer of the base $\psi^{(i)}_{\text{NN}}$ for ticker i contained a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. We froze W and added a rank-r perturbation $\Delta W = \frac{\alpha}{r} BA$ with $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, and a scalar scaling

α Hu et al. (2022). We initialized A from a small Gaussian and set $B = 0$, so $\Delta W = 0$ at the start of training and the adapter weights moved away from zero only as training reduced the day's loss. We trained one adapter per (ticker, capture-date) cell with rank $r = 2$ and $\alpha = 2$ across the three Dense layers, jointly with a scalar update $\Delta \log \theta_i$ on the per-ticker level; the architecture mirrored the base $\psi_{\mathrm{NN}}^{(i)}$ exactly so the adapted forward pass was

$$\sigma^2_{\mathrm{model}}(K, T, t) = e^{\log \theta_i + \Delta \log \theta_i} \cdot \psi_{\mathrm{NN}}^{(i),\,\mathrm{LoRA}}\big(\ln T,\, \ln(K/S_t);\, W + \tfrac{\alpha}{r} BA\big). \tag{14}$$

For the operating per-ticker architecture ($2 \to 16 \to 16 \to 1$ Dense with tanh activations), the rank-2 adapter carried 135 trainable parameters per cell: $\{A_\ell, B_\ell\}_{\ell=1,2,3}$ with sizes $(r \times d_{\mathrm{in}}^{(\ell)}) + (d_{\mathrm{out}}^{(\ell)} \times r)$, plus the one-scalar level update. By contrast, the base $\psi_{\mathrm{NN}}^{(i)}$ carried 369 parameters, and a per-day SABR baseline fits three parameters per maturity bucket. The adapter trained for up to 1,500 epochs of Adam against the same MSE-against-market-IV loss in (13), restricted to the day's ladder slice (typically 200–1,000 rows), with a step-decay learning-rate schedule and patience-150 early stopping; convergence was typically reached by epoch 300–600. The forward-simulation pipeline of (2) was unaffected by the adapter: the base $\psi_{\mathrm{NN}}^{(i)}$ remained the calibration target for the multi-date ladder corpus, and the adapter served only as a mark-to-market overlay applied at the day of contract entry. Forward variance paths continued to use the unadapted base $\bar{\theta}_i$ and $\psi_{\mathrm{NN}}^{(i)}$ exactly as before.

3.5 Streaming refit trigger

A daily refit on every (ticker, date) cell would defeat the framework's compute argument and reintroduce the SABR-style independent-slice failure mode. We therefore used a streaming trigger that decided online whether to fit an adapter for ticker i on date t. The trigger fired when both an absolute and a relative condition held: today's base-vs-market RMSE on the ticker's slice exceeded an absolute floor τ_{abs} (10% IV in our experiments), and its robust z-score over the trailing four *non-flagged* dates exceeded a relative threshold τ_z. The robust z-score was

$$z_t \;=\; \frac{\mathrm{RMSE}_t \;-\; \mathrm{median}(\mathcal{T}_t)}{1.4826 \cdot \mathrm{MAD}(\mathcal{T}_t)}, \tag{15}$$

with $\mathcal{T}_t$ the most recent four dates on which the trigger did *not* fire. Excluding flagged days from the baseline was essential: in pilot runs with a plain mean and standard deviation, the post-INTC-print 04-23 blowup contaminated the trailing window and hid the more moderate 04-28 LLY regime mismatch.

4 Calibration and Validation

4.1 Sector-specific neural calibration on the 31-ticker ladder

The parametric ψ of equation (6) provided the analytic starting point but assumed a single global smile and term-structure shape across all tickers, an assumption that became a bottleneck on a heterogeneous universe spanning sectors with structurally different risk profiles. To quantify the gap and the value of relaxing it, we collected full option ladders for 31 tickers spanning six sectors on fifteen capture dates between 2026-04-14 and 2026-05-11, producing 234,549 observations after filtering to moneyness $K/S \in [0.80, 1.20]$, non-zero bids, and IV $\in (0.01, 2.0)$. The sector partition followed conventional GICS-style groupings (Supplementary Table S1): technology (10 tickers), healthcare (8), financials (4), energy (3), retail (3), and broad-market ETFs (3). The corpus carried systematic cross-sector variation: mega-cap ETFs exhibited shallow smiles with pronounced term structure, technology names carried elevated near-term skew from earnings catalysts, healthcare names showed idiosyncratic bumps around trial and FDA dates, and commodity-linked energy tickers displayed skew patterns shaped by supply shocks rather than equity crash risk. Against this partition, we quantified the value of relaxing the global-shape assumption by fitting three nested models on the pooled ladder using a common loss (mean squared IV error) and a common level parameterization (one θ_{ticker} per ticker), varying only the representation of ψ: a global parametric ψ with the five β-parameters shared across all tickers, a global neural ψ_{NN} shared across all tickers that replaced the parametric form with the neural architecture introduced in the methods, and a sector-specific neural $\psi_{\text{NN}}^{(c)}$ with one network per sector. This design isolated the contribution of model capacity (parametric $\to$ NN) from the contribution of sharing granularity (global $\to$ sector).

We compared the three nested models on the pooled corpus and found that both axes contributed independently (Table 1): moving from the global parametric ψ to the global neural ψ reduced overall RMSE from 12.48% to 11.47% IV (an 8.1% relative improvement) by accommodating smile shapes the log-polynomial form could not express, and moving from the global to the sector-specific ψ_{NN} further reduced RMSE to 10.24% (a 17.9% cumulative relative improvement over the parametric baseline) by letting within-sector shapes specialize away from the global average. A per-day Hagan-2002 SABR reference fitted independently to each (date, ticker, maturity-bucket) slice produced a pooled RMSE of 9.05% IV across the same 234,549 observations; that figure undercuts the sector-NN ψ by 1.2% IV, which we read as the cost of cross-date pooling that lets ψ serve forward simulation. SABR pays for its tighter pooled fit by refitting $\sim$6,072 parameters across the corpus (one (α, ρ, ν) triple per slice with $\beta = 0.5$ fixed), and the resulting calibration is a snapshot of each (date, ticker, expiry) cross-section that cannot price an unseen date without re-fit, propagate forward under stochastic dynamics, or carry a shared shape across tickers in a sector; the sector ψ_{NN} encodes a single calibrated surface (369 parameters per sector, six sectors) reusable across all 15 dates and at simulation time. These figures were in-sample fits to the pooled corpus; a leave-one-date-out check that held out the latest capture and retrained on the remaining fourteen days produced a pooled test RMSE of 10.89% against a train RMSE of 10.21%, leaving a generalization gap of +0.68% IV (Supplementary Table S4) and confirming that the sector-NN result reflected faithful generalization rather than overfit to the multi-date pool. We further audited the calibrated ψ surfaces against the static no-arbitrage conditions $\partial^2 C/\partial K^2 \geq 0$ and $\partial C/\partial T \geq 0$ on a 20×20 moneyness-maturity grid for six sector surfaces (one representative ticker per sector) and the two per-ticker GS and LLY surfaces used in the forward scenarios below; across the eight surfaces, 9.8% of grid points (281/2880) violated the butterfly condition and 0.8% (23/3040) violated the calendar condition, concentrated in the deep wings where ladder coverage was thinnest (Supplementary Table S6). We report these rates as a known limitation of the unconstrained neural surrogate; introducing no-arbitrage penalties in the calibration loss is a natural extension reserved for future work.

Table 1: Overall RMSE comparison on the 31-ticker ladder (234,549 observations across fifteen capture dates). The parametric baseline fit a single global set of five β parameters; the shared NN replaced the parametric ψ with one global network; the sector NN used one network per sector. All three models fitted one θ_{ticker} per ticker jointly with the shape parameters. The per-day SABR reference fitted Hagan-2002 SABR ($\beta = 0.5$, free parameters α, ρ, ν) independently to every (date, ticker, maturity-bucket) slice and pooled the residuals (2,024 separate fits, $\sim$6,072 free parameters in aggregate); it shares no parameters across dates and provides no forward-generation mechanism, so its pooled RMSE is the practitioner reference against which the cross-date pooling cost of ψ is measured rather than a peer model.

Model	RMSE (% IV)	Improvement vs. parametric
Parametric (5 β, global)	12.48	(baseline)
Neural ψ (1 network, global)	11.47	8.1%
Neural ψ (6 networks, sector)	10.24	17.9%
Per-day SABR (per slice)	9.05	27.5%

The per-sector decomposition exposed where each incremental model capacity helped most (Supplementary Table S2). The global NN uniformly improved every sector over the parametric baseline, with the largest absolute gains in ETF (10.54% $\rightarrow$ 8.69%) and retail (11.73% $\rightarrow$ 10.23%), where the parametric log-polynomial struggled to fit shallow ETF smiles and idiosyncratic retail patterns simultaneously. Moving to sector-specific networks delivered a further 1 to 3% IV reduction in ETF, financials, retail, and technology but barely moved the needle in healthcare (12.71 $\rightarrow$ 12.53%) and energy (8.24 $\rightarrow$ 7.88%), reflecting cross-ticker dispersion in those sectors that a single shared ψ could not absorb even when freed from the parametric form. Technology remained the hardest fit overall (14.47% with sector NN); inspection of per-ticker residuals (Supplementary Table S3) traced this to heterogeneity within the sector itself rather than to a failure of the sector NN as a whole: within technology, names such as GOOG (9.57%), AVGO (9.94%), and NVDA (10.14%) fit comfortably under the shared Tech surface, while QCOM (21.46%), MU (18.45%), INTC (17.78%), and AMD (17.74%) pulled the sector RMSE upward. The pattern was consistent with these outlier tickers carrying idiosyncratic smile shapes (elevated wings, asymmetric skew) that a shared Tech ψ could not simultaneously accommodate without degrading the fit of the betterbehaved names. The same story repeated in healthcare, where ABBV (9.13%) and UNH (10.39%) fit well but PFE (17.66%), MRNA (17.49%), and AMGN (13.36%) drove the sector average. The per-ticker θ_{ticker} values recovered a sensible IV level ranking: high-variance names with concentrated risk (INTC, MRNA, MU, AMD) occupied the top of the sorted order at 94 to 117% baseline IV, large-cap diversified names (AAPL, JPM, WMT) clustered near the middle at 46 to 52%, and the broad-market ETFs (SPY, QQQ, IWM) sat at the low-variance end near 35 to 42%. Within-sector clustering was evident in the sorted level estimates, confirming that sector membership captured a dominant axis of the cross-sectional IV structure.

Beyond the aggregate RMSE comparison, the residual structure under the sector NN was concentrated in the wings (moneyness below 0.90 and above 1.10) rather than near the money, consistent with the in-sample fits of the single-ticker baselines and with the known difficulty of pricing deep-OTM contracts where liquidity is thin and quoted IV is noisier. The learned ψ surfaces were smooth in both $\ln \tau$ and $\ln m$ and displayed the expected qualitative features, including put skew (ψ rising for moneyness below unity), smile curvature in the wings, and a mild term-structure gradient. The sector-to-sector differences in these surfaces motivated relaxing the global-shape assumption in the first place: the healthcare and technology surfaces curved more sharply in the near-term wings than the ETF surface, and the energy surface displayed a subtle asymmetry consistent with the commodity-linked risk profile of the sector. This calibration established the sector-specific neural ψ as the operating point of the framework on the pooled in-sample corpus, but the multiplicative decomposition supported a finer relaxation: one ψ_{NN} per ticker. We took up that extension on the same corpus below, where per-name sample sizes were large enough to fit per-ticker networks of the same architecture on nearly the full universe.

4.2 Per-ticker neural calibration on the pooled corpus

We trained one ψ_{NN} per qualified ticker on the full pooled corpus and recovered substantial within-sector shape detail that the sector network could not express (Fig. 1). Twenty-nine of the thirty-one tickers cleared a 2,000-observation threshold sufficient to fit a ψ_{NN} of the same architectural family as the sector level; the two unqualified tickers (PFE at 1,300 observations, BMY at 1,847) were held on the sector ψ_{NN} as a fallback. Network width followed sample size: tickers with at least 5,000 observations received the sector-NN $2 \to 16 \to 16 \to 1$ architecture, smaller-N qualified tickers used $2 \to 8 \to 8 \to 1$. The per-ticker ψ_{NN} reduced the pooled in-sample RMSE from 10.24% under the sector NN to 9.73% under the combined model (a 5.0% relative reduction) and improved twenty-one of twenty-nine qualified tickers, with the largest absolute gains on JNJ (−3.97%), MRNA (−3.76%), AAPL (−1.71%), and INTC (−1.49%)—all in sectors with the most heterogeneous within-sector smile geometry. The shape detail recovered by per-ticker fits (steeper put skew on NVDA than the shared Tech surface, gentler near-the-money curvature on MSFT, the upward right wing on LLY missing from the larger-cap-dominated Healthcare surface) is visible in Fig. 1.

We translated the IV-RMSE story into a dollar-pricing story on the most recent capture (2026-05-11) by repricing each panel via 200-step CRR American at the IV implied by each calibration

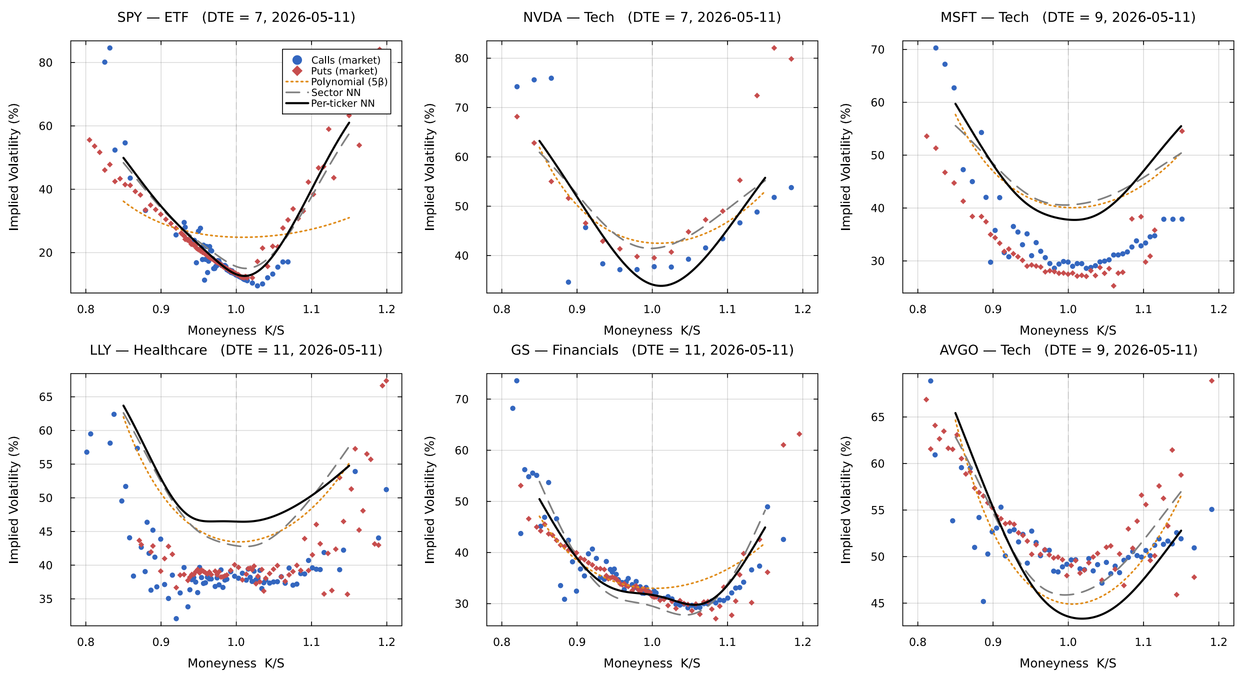

Figure 1: Per-ticker ψ_{NN} smile fits on the fifteen-date pooled corpus. Representative IV smile panels at mid-DTE for six qualified tickers spanning four sectors (SPY, NVDA, MSFT, LLY, GS, AVGO). The per-ticker network (solid black) tracked the smile shape more tightly than the sector network (dashed gray) on every panel, with the largest gap on NVDA and MSFT where the shared Tech surface flattened to accommodate AVGO's elevated wings.

tier, and observed that the per-ticker ψ_{NN} collapsed dollar errors to within \$1 of market mid on liquid names while LLY carried a uniform IV-regime bias (Fig. 2). On SPY (DTE 7) and GS (DTE 11), the per-ticker ψ_{NN} sat inside the local bid-ask ribbon for 33% and 88% of contracts respectively, dominating both the parametric and sector tiers; on NVDA (DTE 7), all three tiers clustered within $\pm$\$0.7. LLY (DTE 11) showed a uniform positive bias of +\$2 to +\$6 across every model tier: the per-ticker ψ_{NN} encoded the time-averaged LLY IV across the fifteen capture dates, including the 04-28 drawdown day, so on a calmer 05-11 the static ψ overestimated IV by several percentage points and the dollar error followed. The market-IV reference (yfinance IV repriced through CRR) sat at $\sim$ \$0.6 on LLY, confirming the CRR pipeline was sound and the residual was an IV-regime mismatch rather than a pricing error—a failure mode the low-rank daily adapter subsequently addresses.

Within the qualified set, eight tickers (BAC, JPM, WFC, CVX, OXY, XOM, UPS, WMT) did not gain over the sector NN; these were concentrated in financials, energy, and retail, sectors whose sector NN had already absorbed most of the within-sector smile dispersion. The temporal-generalization study below used an earlier eight-day window from the same capture (2026-04-14 through 04-24) with a fixed train/test split, so the holdout block was held apart from the pooled in-sample fits.

4.3 Temporal holdout and earnings-aware calibration

The pooled in-sample fit established that the sector-specific ψ_{NN} could absorb the cross-sectional smile geometry of the 31-ticker universe, but it left the temporal-generalization question unanswered: would a ψ trained on one block of capture days price contracts on a held-out subsequent block? To answer this, we used an eight-day subset of the pooled capture (2026-04-14 through 04-17 and 04-21 through 04-24, comprising 130,730 filtered observations) and split it into a six-day training block (04-14 through 04-22, 103,897 observations) and a two-day temporal holdout (04-23 and 04-24, 26,833 observations). The same 31 tickers appeared on every capture day, so the per-ticker θ_i learned on the training block transferred directly to the holdout. We then asked a sharper question: which fraction of any test-set generalization gap was attributable to scheduled corporate events embedded in the holdout window rather than to architectural overfit? To separate these contributions, we ran the sector ψ_{NN} under three configurations: (A) a pooled control identical to the in-sample architecture above (two inputs $\ln \tau$, $\ln m$, no exclusion); (B) a non-earnings baseline that excluded all training and test rows for which the row's ticker or any same-sector peer was within ± 3 days of an earnings print; and (C) the earnings-aware model introduced in the methods that fed ψ four inputs ($\ln \tau$, $\ln m$, e, e^{peer}) and used the same train and test rows as the control. The peer feature carried sector-coupling information directly: for any ticker i on day t, e^{peer} collapsed to zero if any same-sector equity printed earnings that day, regardless of i's own distance to its next print.

Table 2: Temporal-holdout sector-NN comparison across four configurations on the same train (04-14..04-22, 103,897 obs) and test (04-23..04-24, 26,833 obs) split. The pooled control (A) reproduced the existing four-day reference number (13.03% test RMSE). Excluding earnings-window observations from both train and test (B) reduced the test RMSE to 7.96% and collapsed the generalization gap from +4.99 to −0.06% IV. Adding the earnings indicator and peer-coupling feature to the input vector (C) on the original train and test rows reduced the pooled test RMSE to 11.98% and the generalization gap to +4.36%. The Horvath-style direct neural IV baseline (D) replaced the $\theta_i \cdot \psi$ decomposition with a single MLP whose input vector additionally carried a 31-ticker 8-dimensional embedding (1,753 total parameters); on the same train/test split it landed at 12.36% test RMSE, intermediate between the 2-input pooled control and the 4-input earnings-aware sector NN.

Configuration	N_{tr}	N_{te}	Train (%)	Test (%)	Gap (%)
A: pooled (2-input)	103,897	26,833	8.05	13.03	+4.99
B: non-earnings (2-input)	29,383	4,516	8.02	7.96	−0.06
C: earnings-aware (4-input)	103,897	26,833	7.61	11.98	+4.36
D: Horvath direct (4-input + 8-d ticker embedding)	103,897	26,833	8.29	12.36	+4.07

The three-way comparison resolved the generalization question (Table 2). Configuration B revealed that scheduled earnings events accounted for the entire test-time generalization gap of the sector NN: with all earnings-window observations removed from both splits, the test RMSE (7.96%) fell to within rounding of the train RMSE (8.02%), giving a generalization gap of −0.06% IV. The architecture itself generalized faithfully out of sample; what failed was the model's ability to price options on days when an earnings catalyst dominated the IV surface. Configuration C confirmed that adding event-aware inputs to the shape network recovered roughly one-fifth of the lost performance: relative to the pooled control, the test RMSE dropped from 13.03% to 11.98% (1.05% IV absolute, 8.1% relative), and the generalization gap narrowed from +4.99 to +4.36%. The peer-coupling feature did the heavy lifting in this reduction: the per-ticker breakdown showed the largest improvements on Tech names whose own earnings prints fell outside the immediate holdout window but whose sector peers printed during it (Supplementary Table S7). QCOM (own print 6 days after the holdout) and AMD (12 days after) both benefited from the peer-coupling signal because INTC printed inside the window and dragged their IV surfaces with a $\sim$ 4% IV reduction in test error each. The same mechanism reduced META, MSFT, and GOOG test RMSE by 2 to 4% IV apiece. INTC itself, the print-day ticker, was the one Tech name whose RMSE rose under the earnings-aware fit (33.17 $\rightarrow$ 35.94%); its outsized +2192% EPS surprise produced an unanticipated post-print IV crush that no scheduled-event indicator could predict from the calendar alone.

A direct-neural baseline that replaced the $\theta_i \cdot \psi$ decomposition with a single MLP carrying a ticker embedding (Configuration D, 12.36% test RMSE) landed between the 2-input pooled control and the 4-input earnings-aware fit, confirming that the explicit per-ticker level prior in the $\theta_i \cdot \psi$ form contributes a measurable but modest inductive bias rather than the dominant share of the model's accuracy: a comparable-capacity direct net learned an effective level representation from the ticker embedding itself, and the residual gap to Configuration C (-0.38% test RMSE) reflects the structural advantage of the level/shape factorization rather than a capacity ceiling. These results set the limit of what a calendar-driven event indicator can and cannot do. The limit was sharp: the indicator captured the part of event-day IV behaviour known in advance from the calendar (pre-earnings IV expansion on the reporting ticker, sympathy moves on same-sector peers), but it carried no information about the post-event direction or size of the IV move when the actual print deviated from consensus. The non-earnings baseline of Configuration B was therefore the right reference point against which to compare future event mechanisms: any extension of the model aimed at pricing event-day IV surfaces had to be evaluated against the 7.96% non-event RMSE rather than against the pooled 13.03%, because that gap quantified the residual event signal the current architecture could not reach. We return to this distinction in the discussion, where the failure of the indicator on INTC and its success on the sympathy peers motivate a per-ticker tail mechanism reserved for future work.

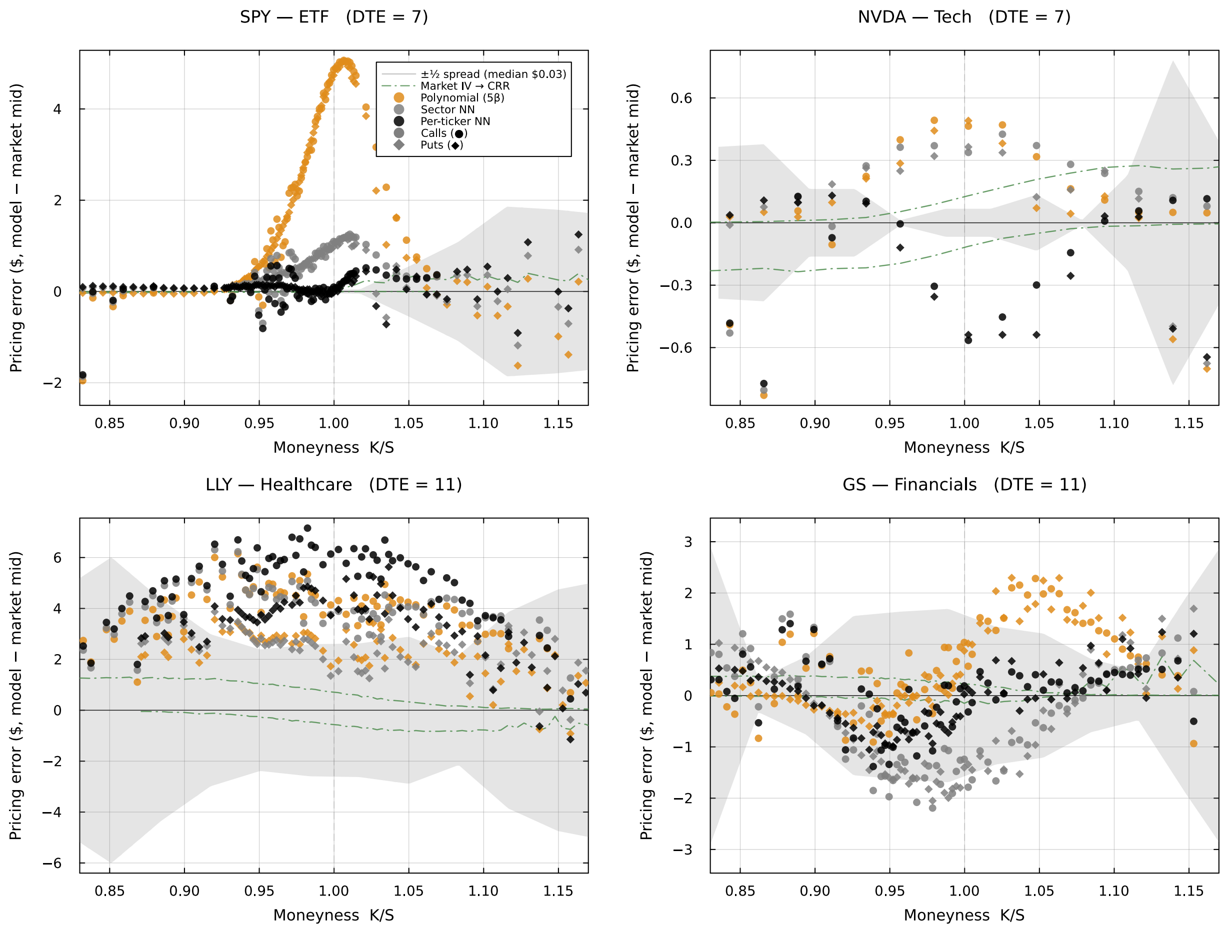

Figure 2: Model-vs-market dollar pricing error (CRR American at the calibrated IV minus market mid) versus moneyness for SPY (DTE 7), NVDA (DTE 7), LLY (DTE 11), and GS (DTE 11) on the 2026-05-11 capture. Four model tiers are overlaid on each panel: parametric ψ (light), sector ψ_{NN} (medium), per-ticker ψ_{NN} (dark), and market-IV reference (green dot-dash; yfinance IV repriced through CRR, near zero by construction). The gray ribbon was the local $\pm\frac{1}{2}$ bid–ask spread band, computed per panel by partitioning the contracts into twelve equal-width moneyness bins, taking the median half-spread (ask − bid)/2 within each bin, and linearly interpolating the twelve bin values onto a fine moneyness grid. A point falling inside the band was a model error smaller than the typical bid–ask spread at that moneyness, i.e., a price within the range the market itself was quoting; the band widened toward the wings because bid–ask spreads were systematically larger for deep in- and out-of-the-money contracts. The per-ticker fit dominated parametric and sector tiers on SPY and GS, matched them on NVDA, and showed a uniform positive bias on LLY consistent with the per-ticker ψ encoding the time-averaged IV across fifteen capture dates and the sampled day (05-11) being calmer than the calibration-window average.

5 Forward-simulated short-premium scenarios

We forward-simulated 1,000 Goldman Sachs (GS) price paths over a 31 trading-day horizon and observed that the model's $t = 0$ Leisen-Reimer fair value sat essentially at the market mid on both legs (put: \$17.26 vs. \$16.51, edge +\$0.75; call: \$16.07 vs. \$16.09, edge −\$0.02; Supplementary Table S8). The two contracts were real GS options pulled directly from the 2026-04-28 ladder capture: a short put at $K_p = \$890$ (market $\Delta = -0.295$, mid \$16.51, market IV 31.3%) and a short call at $K_c = \$970$ (market $\Delta = +0.328$, mid \$16.09, market IV 28.9%), both expiring 2026-05-29 with the underlying spot at $S_0 = \$926.38$. Each path drew from the JumpHMM marginal pretrained on GS returns. Conditional on the path, we evolved a per-contract Cox-Ingersoll-Ross variance v_t with mean-reversion $\kappa = 15.0$, vol-of-vol $\sigma_v = 0.5$, and leverage coupling $\rho = -0.6$ to the per-contract target $\theta_i \cdot \psi_{\text{NN}}((K/S_t)_{\text{std}}, (T - t)_{\text{std}})$, in which the per-ticker neural smile/term-structure surface entered through the drift target so the put and call each followed their own variance trajectory under a shared Brownian shock. At every trading day along every path we re-priced both contracts via 201-step Leisen-Reimer American with $\sigma_t^2(K, S_t) = v_t$, the contract's variance state from the SDE. We adopted Leisen-Reimer over the standard CRR lattice because the Peizer-Pratt construction places the strike K at a fixed lattice position by design at every (S, σ) configuration, which gave smooth Greeks via finite differences without the staircase aliasing that CRR's recombining tree produced when bumps moved K across node boundaries. The seller received the market mid as the entry premium, and the model marked the position to its Leisen-Reimer fair value at every subsequent step; we assumed no scheduled earnings event during the 31-day window so the only sources of variation were diffusion, jumps, and the variance dynamics themselves. With the entry edge near zero, the simulated short was being sold at par with the model's own fair value at the entry capture and the realised path-conditional P&L could be read directly as a return.

The forward-simulation engine is summarised in Algorithm 1. Each simulated trading day along each path produced a (S_t, σ_t) pair from which the LR pricer returned a fair value for the contract; the same (S_t, σ_t) pair fed Supplementary Algorithm 2 for the central finite-difference Greeks. The two algorithms shared the lattice constructor and the per-ticker ψ_{NN} surface, so the Greek bumps inherited the same shape function used for the base mark and did not require a separate calibration. For the scenarios reported in this section, we operated the variance process at the calibration-collapsed level $\bar{\theta}_i := \theta_i$ from (5), with the regime-state and aggregate-mood multipliers in the full θ mechanism of (4) held inactive ($s_t \equiv s_0$, $\gamma = 0$); this isolates the contribution of the per-ticker ψ_{NN} surface to path-conditional pricing from the state-and-mood coupling that the framework also supports.

We rendered the share-price ensemble alongside the short-put and short-call premium ensembles (Fig. 3). On the worst-5% paths (red, top row), GS drifted toward and through K_p, the short-put premium climbed many-fold above its entry price, and the position closed with a mean P&L of −\$113.66 on those paths. On the top 1–5% paths (green, bottom row; the wildest 1% were excluded for legibility), GS pushed past K_c and the short-call premium spiked despite the calendar pulling

Algorithm 1 Forward simulation of path-conditional option premia

Require: JumpHMM marginal for ticker i; per-ticker $\psi_{\mathrm{NN}}^{(i)}$ and calibration-collapsed level $\bar{\theta}_i := \theta_i$ from (5); contract (K, T); CIR parameters (κ, σ_v, ρ); spot S_0; risk-free rate r; number of paths N_p; LR tree depth N_{LR}; step $\Delta t = 1/252$.

Ensure: $\{S_t^{(j)}, v_t^{(j)}, \sigma_t^{(j)}, P_t^{(j)}\}$ for $j = 1, \ldots, N_p$ and $t = 0, \ldots, T$.

▷ Operates at $\bar{\theta}_i$ from (5); $s_t \equiv s_0$ and $\gamma = 0$ in this section

for $j = 1$ **to** N_p **do**

  Draw price path $\{S_t^{(j)}\}_{t=0}^{T}$ from the JumpHMM marginal

  $v_0^{(j)} \leftarrow \bar{\theta}_i \cdot \psi_{\mathrm{NN}}^{(i)}\big(\ln(K/S_0),\, \ln T\big)$ ▷ per-contract entry, eq. (7)

  for $t = 0$ **to** $T - 1$ **do**

    $\sigma_t^{(j)} \leftarrow \sqrt{v_t^{(j)}}$ ▷ per-contract IV, eq. (8)

    $P_t^{(j)} \leftarrow \mathrm{LR}_{N_{\mathrm{LR}}}\big(S_t^{(j)},\, K,\, \sigma_t^{(j)},\, T - t,\, r\big)$ ▷ Leisen-Reimer American

    Draw $Z_S^{(j)}$ from the JumpHMM return innovation; draw $Z_\perp \sim \mathcal{N}(0, 1)$ independent

    $Z_v^{(j)} \leftarrow \rho\, Z_S^{(j)} + \sqrt{1 - \rho^2}\, Z_\perp$ ▷ leverage coupling

    $\theta_t^{(j)} \leftarrow \bar{\theta}_i \cdot \psi_{\mathrm{NN}}^{(i)}\big(\ln(K/S_t^{(j)}),\, \ln(T - t)\big)$ ▷ per-contract drift target

    $v_{t+1}^{(j)} \leftarrow \big|v_t^{(j)} + \kappa(\theta_t^{(j)} - v_t^{(j)})\,\Delta t + \sigma_v \sqrt{\max(v_t^{(j)},\, 0)}\,\sqrt{\Delta t}\, Z_v^{(j)}\big|$

  end for

  $P_T^{(j)} \leftarrow \max(\phi(S_T^{(j)} - K),\, 0)$ ▷ $\phi = +1$ call, $\phi = -1$ put

end for

return $\{S_t^{(j)}, v_t^{(j)}, \sigma_t^{(j)}, P_t^{(j)}\}$

extrinsic value toward zero; mean P&L on the top-5% paths was $-\$176.75$, $1.55\times$ the mean loss on the worst put paths. On both legs the median path stayed inside the 25–75% band and closed flat at the premium received; the short-premium trade made money on the typical path, with both legs carrying a long left tail.

The short-call leg lost asymmetrically because IV at the contract strike rose as $t \to T$ on every path (Fig. 4). The rise was not a model artefact: it was the empirical rise of the smile wings at short maturities that the per-ticker ψ_{NN} learned from the ladder corpus. As $(T - t)$ shrank at fixed $\log(K/S)$, the standardised moneyness moved further out into the wings where ψ_{NN} grew monotonically, and a Gaussian diffusion would have underpriced that wing without the inflation, since short-horizon returns carried much heavier tails than long-horizon returns, which were close to Gaussian by central-limit averaging. The variance v_t tracked the rising drift target $\bar{\theta}_i \cdot \psi_{\mathrm{NN}}(\cdot)$ via mean-reversion, so the IV rise into expiry was inherited directly from the term-structure curvature of the per-ticker ψ_{NN} rather than from a separate level component. On the green tail S_t pushed away from the call strike, pulling the standardised moneyness further into the wing and amplifying the rise; on the red tail the leverage coupling lifted v_t at the same time as S_t fell, so the two effects added.

We decomposed the position risk into Δ, Γ, and Vega along the same paths (Supplementary Fig. S1), computed by central finite differences on the same 201-step Leisen-Reimer American pricer used for the base marks (spot bumps of $\pm 1.5\%$ for Δ and Γ; σ bumps of ± 0.005 for Vega-per-1%-

IV). The long-put Δ began near -0.30 and bifurcated: on the worst-5% paths it ran to -1 as the put pinned ITM at expiry, while on the rest it ran to 0; the long-call mirror image held at the call strike on the top tail. Γ collapsed to zero on every path that drifted away from its strike but spiked sharply on paths that pinned to the strike near expiry, the standard reason traders avoid carrying short-premium positions through expiry. Vega decayed monotonically toward zero on every path (the $\sqrt{T-t}$ factor dominates), but the worst-5% put paths and top-5% call paths held materially more vega than the median through about $t = 20$, exposing the short-premium seller to extra IV-fluctuation risk on the same paths that were already losing on Δ. The signs of the position-level Greeks for the short trader are the negatives of the long Greeks shown.

Both legs showed the characteristic short-premium asymmetry (Fig. 5; moments in Supplementary Table S8): most paths kept the full premium, and a long left tail held the loss-making paths. The short put kept its full \$16.51 premium on 72.2% of paths and closed with a mean P&L of +\$1.52; the short call kept its full \$16.09 premium on 64.3% of paths and its mean P&L of $-$\$9.00 was net negative, with the worst-case loss reaching $-$\$484.57, 1.50$\times$ the worst put loss and 30$\times$ the premium itself. The asymmetry between the two legs reflected the negative skew that GS's per-ticker ψ_{NN} encodes (higher IV at down-strikes than at equally-far up-strikes) together with the leverage coupling that lifted v_t on down-paths; the put premium already reflected the down-tail risk and earned a fair return for taking it, while the call leg ran into a fatter empirical right tail than its symmetric Black–Scholes premium could cover.

We compared the simulated premium-kept rates with the trader's delta-as-probability heuristic and found them within 3% of the rule's prediction on both legs (Fig. 5; Supplementary Table S8). The heuristic approximates Pr[ITM at expiry] $\approx |\Delta|$ and so predicts premium-kept $\approx 1 - |\Delta|$ for a short held to expiry. The GS put's $|\Delta| = 0.295$ implied a 70.5% rule-predicted retention against our simulated 72.2%, and the call's $|\Delta| = 0.328$ implied 67.2% against our 64.3% (deviations +1.7% and -2.9%). The agreement was a non-trivial sanity check: the rule derives from the Black-Scholes identity $\Pr[\mathrm{ITM}] = N(d_2) \approx N(d_1) = |\Delta|$ at short horizons and near-zero drift, while our terminal probabilities emerged from a composition of the JumpHMM marginal, Heston variance dynamics, per-ticker ψ_{NN}, and Leisen-Reimer American pricer, each calibrated to a different objective (return autocovariance, ladder RMSE, lattice convergence) and none calibrated against the rule. The directional residual—positive on the put and negative on the call—was consistent with the small positive long-run drift anchor that the JumpHMM marginal carried: a slight upward pull lifted paths away from the put strike and toward the call strike, reducing put-side breaches and increasing call-side ones relative to a drift-free reference.

5.1 Cross-ticker validation: LLY

We re-ran the identical pipeline on a real LLY option pair from the same 2026-04-28 capture and observed two things: the qualitative short-premium asymmetry from GS reproduced cleanly on a ticker in a different sector at a higher IV regime, and the model's $t = 0$ fair value sat \$1.35–\$1.80 below the market mid on both legs, a reversal of the near-par GS entry edges. The contract was a short put at $K_p = \$825$ (market $\Delta = -0.303$, mid \$23.30, market IV 44.4%) and a short call at $K_c = \$940$ (market $\Delta = +0.309$, mid \$20.76, market IV 44.0%), both expiring 2026-05-29 with LLY spot $S_0 = \$873.83$; every other framework choice was held identical to the GS scenario.

The terminal P&L distribution (Fig. 6; Supplementary Table S9) reproduced the GS pattern: short put kept full premium on 77.4% of paths, short call on 75.7%, with the short-call worst-case loss again exceeding the worst put loss by 1.64× despite the smaller premium received. The same asymmetric-tail pattern on a different sector and a different jump-state mixture showed that the negative skew came from the framework rather than from ticker-specific calibration. Both LLY legs ran above their delta-as-probability prediction by $\sim 7\%$, roughly double the $\sim 3\%$ residual on GS, and the same gap appeared in dollars as the $-\$1.35/-\$1.80$ entry-edge column of Supplementary Table S9. The mechanism is an IV-regime mismatch: LLY had dropped $\sim 5\%$ on 04-28, so the contracts were captured against an elevated-IV state while the per-ticker ψ_{NN} encoded the time-averaged shape across fifteen capture dates. This particular failure mode—a static surface mis-pricing a single-day regime—motivates the low-rank daily adapter introduced below; we revisit the LLY 04-28 case study there and show that a rank-2 adapter closes the entry-edge gap to $-\$1.15/+\$0.46$ with 135 daily trainable parameters.

5.2 Cross-sector validation across five additional tickers

We re-ran the identical pipeline on five additional tickers from the 31-ticker universe drawn to span ETF (SPY), Energy (XOM), Tech (AMD), low-liquidity Healthcare (PFE, the one ticker that fell below the 2,000-observation per-ticker fit threshold and therefore inherits the Healthcare sector ψ_{NN} surface), and large-cap Healthcare (JNJ), and observed that the qualitative short-premium asymmetry reproduced across all five names (Supplementary Table S10). Strikes for each ticker were pulled from the same 2026-04-28 capture at the closest $\sim 30\Delta$ ~30-DTE pair, the Heston backbone $(\kappa, \sigma_v, \rho) = (15.0, 0.5, -0.6)$ was held identical to the GS and LLY scenarios, and each ticker drew from its own JumpHMM marginal with a ticker-prior CCGR anchor matching its sector long-run return (8%/yr for ETF, Energy, Healthcare; 12%/yr for Tech). The premium-kept rates clustered between 59.3% and 78.5% across all five names—the design 30-delta of each contract anchored each leg near the canonical $1 - |\Delta| \approx 70\%$ premium-retention target—and the worst-call/worst-put loss ratio repeated the GS-LLY asymmetric tail on AMD (3.23×: worst-call $-\$361.75$ vs worst-put $-\$111.84$) and reproduced near unity on SPY (the ETF's balanced moneyness profile produces symmetric tails by construction). PFE, the one ticker falling back to its sector ψ_{NN} rather than a per-ticker fit, sustained the smallest dollar tail losses simply because its premiums were small in absolute dollars, but it preserved the same percentage premium-kept structure as the per-ticker tickers, indicating that the sector-NN fallback is operationally viable for low-liquidity names. JNJ exhibited the LLY-style negative entry edge on the put leg (model fair value $-\$0.98$ below market mid), reflecting the same time-averaged-ψ vs. capture-day-regime gap noted for LLY on 04-28; the same gap closed on JNJ's call leg ($+\$0.45$ above market) because the call sat at a different point on the ψ surface. The reproduction of the short-call asymmetric tail on AMD (a tech name with elevated IV $\sim$ 65% versus GS's $\sim$ 30%) and the appearance of balanced tails on SPY together established that the asymmetry is a property of the per-ticker smile geometry that the framework encodes, not a calibration artefact specific to a single sector or volatility regime.

5.3 Earnings-window scenario: INTC across the 04-22 print

The cross-ticker scenarios above and the GS/LLY headline scenarios all sat in event-free 30-day windows by construction. We close this section by running the framework on its most common practical use case: a forward window that deliberately spans a scheduled earnings print. We anchored the scenario at INTC's 2026-04-13 capture (S_0 = \$65.17) and ran a 32-day forward simulation to the contract's 2026-05-15 expiry, placing INTC's 2026-04-22 print at day $\sim$ 7 of the window. The contract was a 30Δ pair from the same capture: a short put at K_p = \$60 (mid \$3.28, market IV 78.2%) and a short call at K_c = \$75 (mid \$2.62, market IV 73.6%). We chose INTC because it carries the largest pre-print to post-print IV-regime gap in the fifteen-date corpus: its 04-22 earnings produced a +2192% EPS surprise driving a $\sim$ 30% realised spot move from \$65.17 on 04-13 to \$84.54 on 04-28 (the same print isolated by the earnings-aware holdout as the residual signal no calendar-based indicator could absorb).

The simulator produced a terminal P&L distribution dominated by middle-of-the-JumpHMM-marginal trajectories rather than by the realised post-print rally: short put kept full premium on 79.5% of paths and short call on 85.8%, both materially above the $\sim$ 70% retention typical of the same 30Δ design on event-free windows (Supplementary Table S10, with INTC's 85.8% call retention as the highest entry). The model's $t = 0$ Leisen-Reimer fair value sat +\$0.22 on the put and +\$0.34 on the call above market mids, indicating that the framework regarded both contracts as fairly priced from the perspective of its time-averaged surface. The empirical $\sim$ 30% INTC rally would have placed the short call \$9.54 in-the-money at the realised path; the 85.8% retention rate reflects the absence of that path in the JumpHMM marginal rather than a calibration success. This is the framework's clearest single-cell limitation under the static-ψ operating point used throughout this section, and we return to it below in the adaptive recalibration section, where a low-rank daily adapter applied to INTC's 04-23 ladder slice reduces the slice RMSE from 68.6% to 5.97% with 135 trainable parameters.

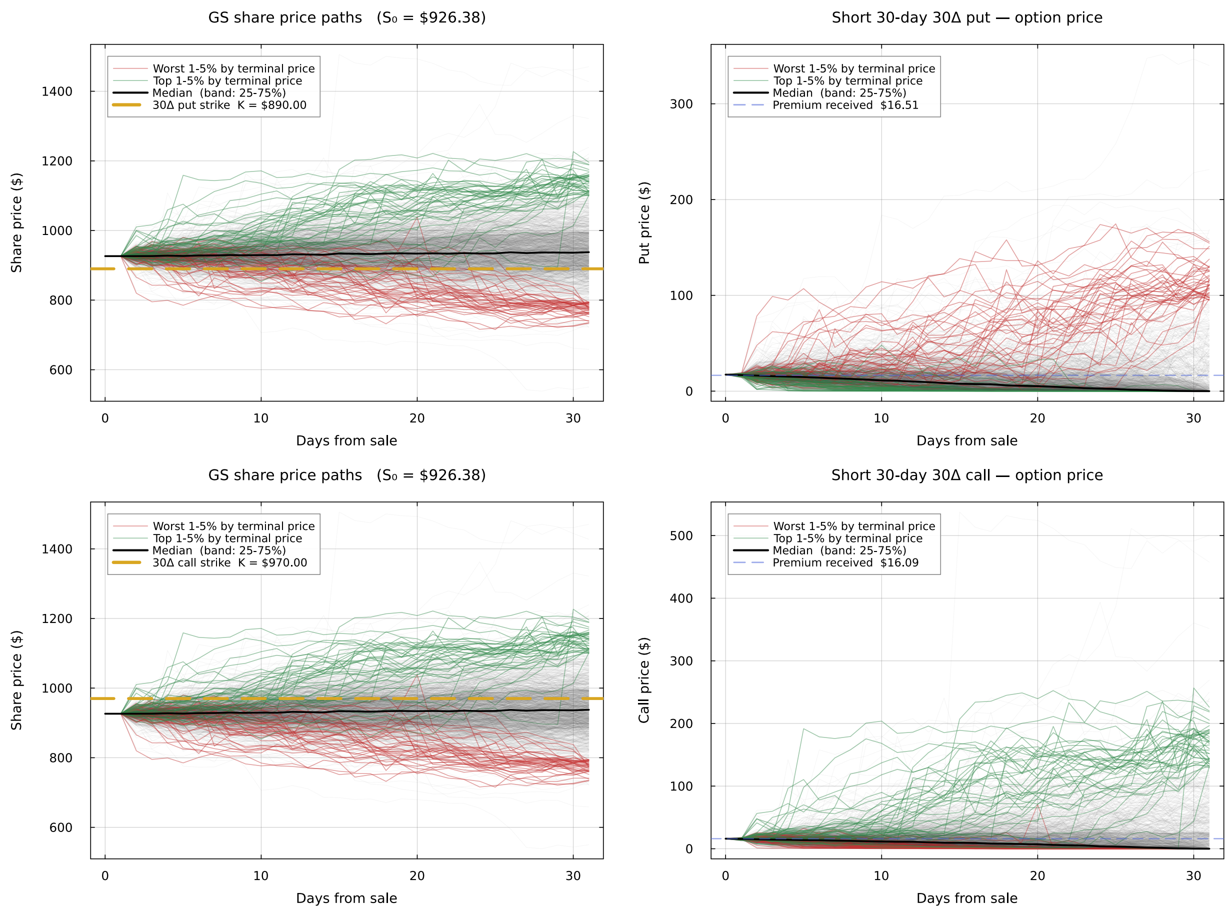

Figure 3: Forward GS share-price paths and short option premium paths for the real GS 2026-05-29 contracts, 1,000 simulated trajectories over 31 days. Top row: share path with $K_p = \$890$ marked (left) and short-put premium (right), with the worst 1–5% by terminal price overlaid in red. Bottom row: share path with $K_c = \$970$ marked (left) and short-call premium (right), with the top 1–5% by terminal price overlaid in green; the most extreme 1% are omitted from the overlay so the figure is not dominated by spike outliers, but the underlying summary statistics use the full top 5%. Black solid: median; shaded band: 25–75% IQR. Blue dashed lines on the option-price panels mark the market mids received at sale (\$16.51 put, \$16.09 call).

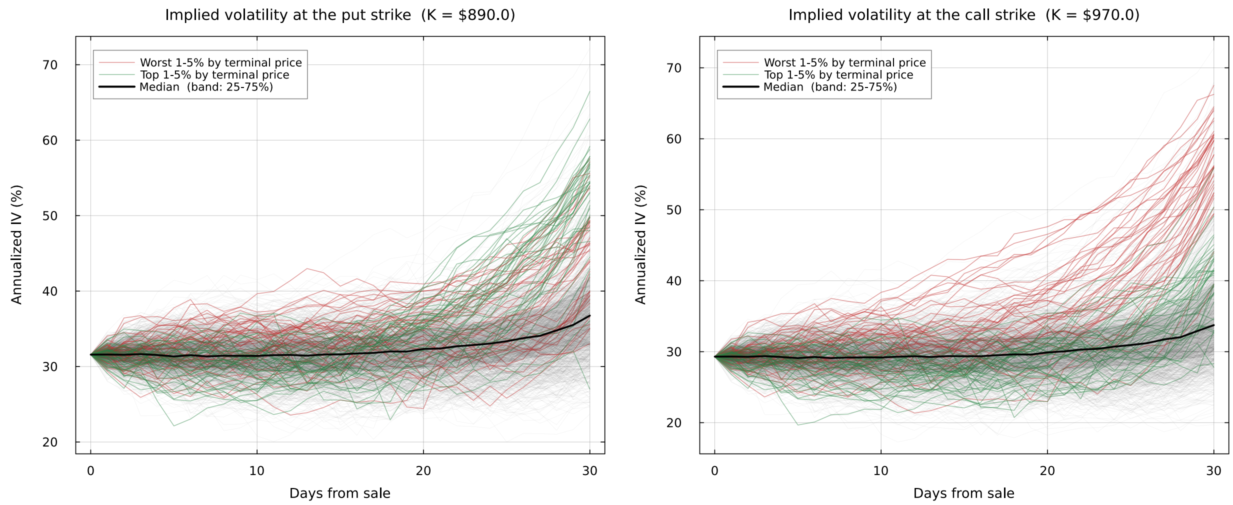

Figure 4: Path-conditional implied volatility at the put strike (left) and call strike (right) over the 31-day horizon for the GS scenario. Both panels showed IV rising into expiry: at fixed $\log(K/S)$, shrinking $(T-t)$ pushed the standardised moneyness deeper into the wing of ψ_{NN}, which the per-ticker network learned to be progressively higher and steeper at short horizons. Tail-binned paths (red, green) showed an additional rise driven by S_t moving away from K and, on the put side, leverage coupling lifting v_t on down-paths.

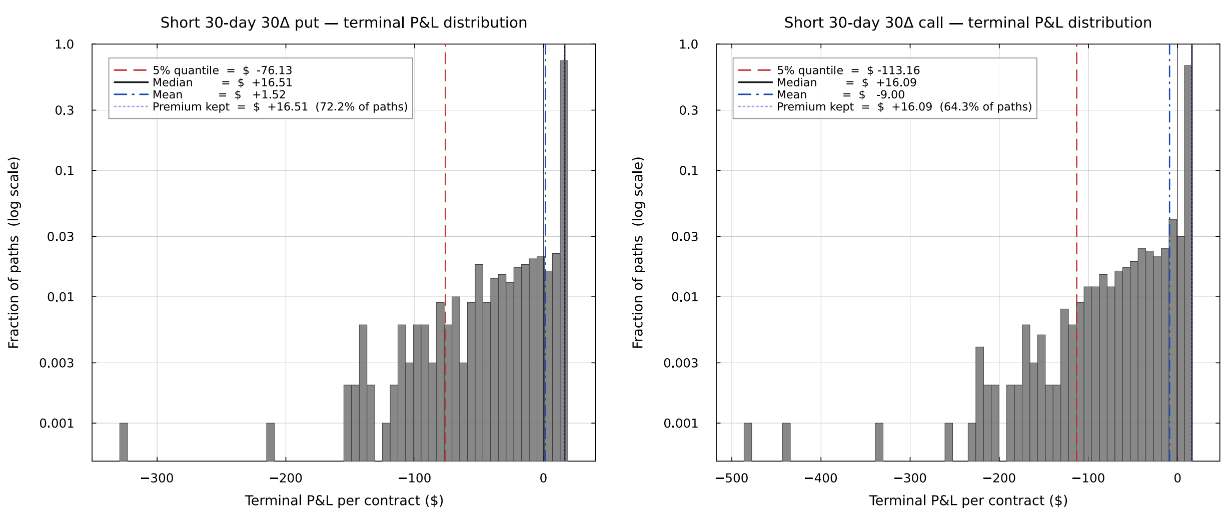

Figure 5: Terminal P&L distributions per contract (1,000 paths, log-y) for the GS scenario. Both legs showed the same short-premium pattern: most paths kept the full premium, and a long left tail held the loss-making paths. The short call's left tail was substantially fatter than the put's despite a smaller premium received.

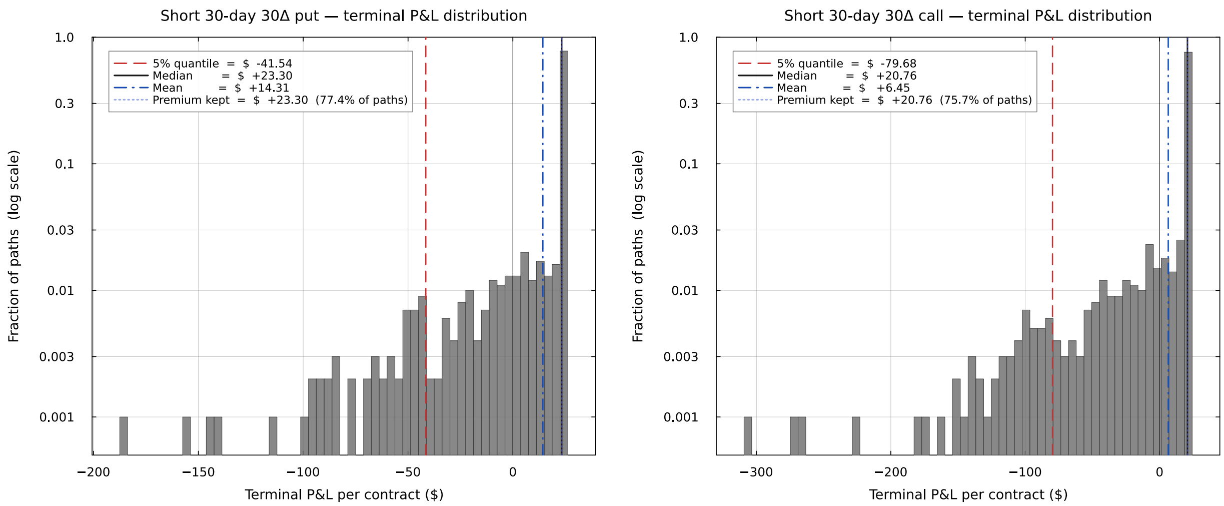

Figure 6: LLY terminal P&L distributions per contract (1,000 paths, log-y), parallel to Fig. 5. Short put kept full premium on 77.4% of paths (mean P&L +\$14.31); short call kept full premium on 75.7% of paths (mean P&L +\$6.45) with a worst-case loss of −\$303.88, 1.64× the worst put loss and 14.6× the premium received. Full statistics in Supplementary Table S9; share-price paths, IV trajectories, and Greek panels in the Supplementary Material.

6 Adaptive recalibration via low-rank daily adapters

The pooled multi-date calibration delivered a single coherent IV surface usable for the forward simulator, but it paid for that coherence on two fronts. A per-day Hagan SABR reference fit independently to every (date, ticker, maturity) slice undercut the pooled sector ψ_{NN} on aggregate RMSE by 1.2% IV (9.05 vs 10.24%; Table 1). On individual capture days whose realised IV regime diverged from the multi-date average, the static surface mis-priced both legs of the same-ticker 30Δ pair in the same direction, producing the $-\$1.35/-\$1.80$ entry edges on LLY's 04-28 capture (Supplementary Table S9). Both costs were structural to the multi-date pooling rather than to the architecture itself. We resolved them by introducing a daily low-rank adapter on top of the frozen base ψ_{NN}, gated by a streaming median-and-MAD refit trigger.

The trigger recovered 74% of the 112 corpus cells exceeding the 12% RMSE refit-needed threshold at the Youden-optimal operating point ($\tau_z = 0$, $\tau_{\mathrm{abs}} = 10\%$) with 67% precision, and 67% at the more conservative $\tau_z = 1.5$ default with 69% precision (Supplementary Table S13; Fig. S7). The flagged dates included INTC's 04-21 pre-print, 04-22 print, 04-23 post-print, and 04-28 regime-echo dates; the trigger missed only the 04-14 capture, which fell inside the four-date warm-up window where no streaming statistic could discriminate without trailing data.

6.1 Empirical sweep across 20 (ticker, date) cells

We tested the adapter on 20 cells drawn deterministically from the corpus: the five highest-base-RMSE cells (problem days), the five lowest (controls), and ten cells from the middle 50% of the RMSE distribution (Table S11). On each cell we compared two adapter variants against the unadapted base: V0 updated only the scalar $\log\theta_i$ (1 parameter), and V1 used the rank-2 LoRA on each Dense layer together with $\Delta\log\theta_i$ (135 parameters). V1 reduced the day-slice RMSE on 20 out of 20 cells (it never overfit) and beat V0 on every cell. The improvement increased with base RMSE: median absolute reduction was 20.4% IV on the five worst cells, 2.4% on the middle ten, and 0.7% on the five controls (Fig. S5A). On the control cells, the adapter barely perturbed the surface: when the base $\psi_{\mathrm{NN}}^{(i)}$ already fit a ticker's day-slice to within 4% IV (the GS cells), the rank-2 adapter shifted the level scalar by less than 0.07 in $\log\theta$ and the shape perturbation norms $\|B_\ell A_\ell\|_F$ all stayed below 2.7. On the worst cells, the middle-layer perturbation $\|B_2 A_2\|_F$ increased steadily with the base RMSE (Fig. S5B), so the adapter put most of its capacity in the middle layer rather than the input or output projection, the same pattern reported for transformer LoRA Hu et al. (2022).

The LLY 04-28 cell appeared as a single row in the sweep. The base $\psi_{\mathrm{NN}}^{(\mathrm{LLY})}$ produced a 15.66% slice RMSE, V0 reduced it to 11.78% by lifting $\log\theta_{\mathrm{LLY}}$ by +0.35, and V1 reduced it to 4.34% with a much smaller level shift (+0.03) and a shape perturbation that lifted the wings without disturbing the near-the-money region. Re-pricing the 30Δ LLY 2026-05-29 put and call from the LLY cross-ticker scenario under each variant produced (Table S12): the base $-\$1.35$ put / $-\$1.80$ call entry edges; V0 overshot to $+\$6.15$ / $+\$5.44$ because the global level lift over-priced the 30Δ

strikes while fitting the wings; V1 closed the gaps to $-\$1.15$ / $+\$0.46$, the call leg landing essentially at par. The same comparison on INTC's 04-23 post-print blowup (the largest single failure in the corpus) gave a 68.6% base RMSE, 17.1% under V0, and 5.97% under V1, with A_ℓ, B_ℓ pattern norms 0.70/2.02/0.68 and $\Delta \log \theta_{\mathrm{INTC}} = +0.30$.

6.2 No-arbitrage diagnostic on adapted surfaces

The rank-2 LoRA adapter carried no structural constraint to preserve butterfly or calendar arbitrage, so we re-ran the static-arbitrage check on the 20 sweep cells under both the base and the V1-adapted ψ_{NN} (Fig. S6). Aggregated across the 20 surfaces, butterfly violations rose from 9.78% to 12.74% of grid points and calendar violations from 1.01% to 2.32%. The amplification was bounded but concentrated on aggressive adapters fit to the highest-RMSE cells: INTC 04-23, AMD 04-23, and QCOM 04-23 each saw butterfly violation rates roughly triple under adaptation, while six of the twenty cells (mostly the GS controls) actually saw violations decrease. The natural extension is a Variant 2 that adds an arbitrage-penalty term to the adapter loss — the LoRA framework imposes no obstacle to such a penalty, and the per-cell adapters could enforce sector-specific weights on butterfly vs calendar consistency — but we did not implement it here. We report the violation rates as a known property of the rank-2 unconstrained adapter, the cost we pay for the slice-RMSE gain documented above.

6.3 Operating regime

The adapter behaved symmetrically across the corpus. Where the multi-date base $\psi_{\mathrm{NN}}^{(i)}$ matched the day's market regime, the rank-2 adapter barely perturbed the surface (sub-percent slice-RMSE improvement, near-zero level shift) and the streaming trigger correctly chose not to refit. Where the day's regime diverged from the calibration-window average—LLY 04-28's elevated-IV regime, INTC 04-22/23's earnings-print and post-print blowup, the residual +4.36% generalization gap of Configuration C in Table 2—the adapter closed most of the gap with about a hundred trainable parameters per cell, the trigger flagged the right days online, and the forward simulator continued to operate on the unadapted base. The adapter is therefore not a daily-refit replacement for the multi-date pooled fit, which would break the forward-simulation property the framework exists to provide; it is a calibration-time overlay that closes the multi-date-vs-daily gap on the subset of days where the base $\psi_{\mathrm{NN}}^{(i)}$ alone is insufficient.

7 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ-function produced realistic IV surfaces from structural return dynamics alone, and the same decomposition of θ into a per-ticker level and a shape function ψ accommodated a hierarchy of representations spanning a parsimonious analytic form, a globally shared neural surrogate, and a sector-specific neural surrogate. The equilibrium initialization $v_0 = \theta(t = 0)$

eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction; this was not merely a computational convenience but a structural constraint that ensured the initial IV surface was fully determined by the θ-function, consistent with the view that IV surfaces reflect the conditional distribution of returns under the risk-neutral measure Cont and da Fonseca (2002). The ψ-function played an analogous role to the SVI parameterization Gatheral (2004) in that it provided a parsimonious representation of the smile and term structure, but with a critical difference: the parameters entered a generative model rather than a static interpolation, so the calibrated surface could evolve dynamically as the underlying HMM state changed.

Scaling the calibration to the 31-ticker fifteen-date ladder revealed the limits of the parametric form as a universal shape function and motivated the hierarchy of neural representations. A single global ψ fit across the six sectors achieved an overall RMSE of 12.48% IV (Table 1), dominated by systematic misfits in sectors whose smile geometry departed from the semiconductor baseline (technology at 15.58%, healthcare at 13.49%; Supplementary Table S2). Replacing the parametric form with a globally shared neural ψ reduced the overall RMSE to 11.47% without changing the sharing granularity, confirming that part of the gap was a functional-form bottleneck in the log-polynomial representation. Loosening the sharing to one network per sector reduced the overall RMSE further to 10.24%, with the sector NN dominating the shared NN in every sector, which established that both axes (capacity and granularity) contributed independently. The remaining error was concentrated in two structural sources: within-sector heterogeneity in technology and healthcare, where specific tickers (QCOM, MU, INTC, AMD, PFE, MRNA) carried idiosyncratic smile shapes not expressible by a shared sector-level ψ, and residual small-sample effects for the two healthcare tickers (PFE at 1,300 observations and BMY at 1,847) that still sat below the 2,000-observation per-ticker training threshold. Both sources are addressable through the same decomposition: the per-ticker roadmap relaxed shape sharing to the ticker level for the data-rich names, and additional capture days continued to reduce the small-sample variance for the remaining thin names.

Extending the corpus to eight capture days and running a temporal holdout reframed the generalization question more sharply than the in-sample RMSE alone could. The pooled-control sector NN trained on six days and evaluated on the next two carried a 4.99% IV generalization gap (8.05% train, 13.03% test; Table 2), which on its face suggested a substantial overfit. Excluding observations within ± 3 days of any same-sector earnings print from both train and test eliminated that gap entirely (8.02% train, 7.96% test, -0.06% generalization gap), revealing that the architecture itself generalized faithfully out of sample and that the entire 4.99% gap was attributable to scheduled corporate events embedded in the holdout window. Adding earnings-distance and same-sector peer-distance features to the shape network closed roughly one fifth of the gap on the original train and test rows (11.98% test, $+4.36\%$ generalization gap). The per-ticker decomposition exposed the mechanism (Supplementary Table S7): the peer-coupling feature carried the IV-spillover signal, so QCOM (own print six days after the window) and AMD (twelve days after) saw $\sim 4\%$ IV reductions in test error from the INTC print on a holdout day, and the same mechanism cut META, MSFT, and GOOG test RMSE by 2 to 4% IV. INTC itself was the only Tech name whose error rose under the earnings-aware fit, driven by an outsized +2192% EPS surprise that the calendar-based indicator could not predict. The boundary was therefore sharp: the indicator captured the part of event-day IV behaviour known in advance from the calendar (pre-print IV expansion on the reporting ticker and sympathy moves on same-sector peers), but it carried no information about the post-print outcome itself when consensus was wrong. The non-earnings baseline of 7.96% test RMSE is the right reference for what the current architecture can do under faithful temporal generalization; the residual gap to the pooled 13.03% quantifies the part of the surface that requires a richer event mechanism than the calendar.

Several limitations bounded the current claims. The cross-sector partition used conventional GICS-style groupings; whether alternative groupings (mega-cap versus semiconductor within technology; pharma versus biotech within healthcare) would improve the sector-NN fit without fragmenting the corpus was an empirical question that additional data would clarify. The forward scenario analysis demonstrated the framework on real 31-DTE 30-delta GS contracts pulled from the 04-28-2026 capture and produced a coherent path-conditional risk decomposition (Leisen-Reimer American Greeks via finite differences, terminal short-premium P&L, IV trajectories, and the entry-edge gap between the model's lattice fair value and the market mid; Figs. 4 and 5; Supplementary Table S8 and Fig. S1). A cross-ticker re-run on LLY (Supplementary Material) confirmed the qualitative pattern on a ticker with a different sector and a higher IV regime; but the validation remained limited to two tickers from a single 30-day window in a single market regime, and broader validation across sectors and volatility environments—together with a re-run of the scenarios under the sector-NN ψ for tickers without per-ticker networks—would strengthen the generalizability claim. The current scenarios also assumed no scheduled earnings event during the simulation window, which the earnings-aware calibration flagged as the dominant residual driver of test-time error; a natural extension would forward-simulate across an earnings print and quantify how the path-conditional Greek profile and terminal P&L shifted when the event-aware ψ fed the same simulation pipeline. The earnings-aware fit recovered the pre-print component of the event-day signal but left the surprise-driven component uncaptured.

A natural extension restored the per-ticker tail term that the simulation pipeline already supported for forward state evolution and that the calibration omitted: replacing the operational θ_i with $\theta_i \cdot (1 + \gamma_i M_i^{(t)})$, where $M_i^{(t)}$ was the JumpHMM tail-state probability for ticker i at time t and γ_i was a learned per-ticker tail sensitivity, would absorb idiosyncratic IV inflation from any source whose return-side imprint already drove the HMM into a tail state, including but not limited to earnings surprises. Restoring this term required state-conditioned histories rather than observation-day surfaces, so it sat naturally at the simulation interface: the calibrated ψ_{NN} remained the IV shape function, the calendar-based earnings inputs (e, e^{peer}) remained the scheduled-event indicator, and $\gamma_i M_i$ became the unscheduled tail mechanism populated from the JumpHMM at simulation time. The natural reference point for evaluating either of these extensions was the non-event 7.96% test RMSE established by the earnings-aware calibration (Table 2), against which any further improvement would represent a genuine reduction in the residual event signal rather than a recovery of cross-sectional fit already accounted for by the sector ψ. Beyond these specific limitations, the primary intended use of this framework was generating realistic synthetic training data for machine-learning models in finance, with applications ranging from training option price predictors and learning delta-hedging policies via reinforcement learning Buehler et al. (2019) to augmenting scarce real option data for transfer learning. The framework naturally supported computation of Greeks (Δ, Γ, $\mathcal{V}$, Θ) via finite differences on the binomial-tree pricer, and the GS scenario showed that the Leisen-Reimer variant produced smooth Greeks at modest tree depth where CRR's recombining lattice exhibited node-snapping aliasing in σ and S bumps. Including these Greeks in the synthetic dataset would support hedging applications. Finally, the current mood mechanism created correlated IV dynamics through a shared stress signal; a richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respected observed correlation patterns between individual-name and index volatility.

8 Conclusion

In this study, we developed a computational pricing pipeline for American options under regime-coupled stochastic volatility, in which the IV-shape function was calibrated once against a historical option-ladder corpus and forward simulation then propagated path-conditional implied volatility without re-ingesting market IV along the simulated path. The pipeline linked a Jump Hidden Markov Model to a modified Heston variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t\!=\!0)$ eliminated the initial IV surface as a free parameter at simulation time, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated the smile and term-structure shape function ψ through a hierarchy of representations on a 31-ticker, fifteen-date, six-sector option ladder. The five-parameter parametric form served as the analytic baseline; replacing it with a globally shared neural surrogate and then with sector-specific neural surrogates reduced the overall fit error by 8% and 18% respectively (Table 1), with independent contributions from added model capacity and loosened sharing granularity. A temporal holdout on an eight-day subset of the same capture further showed that scheduled earnings events accounted for the entire test-time generalization gap of the sector NN, and that adding calendar-derived earnings-distance and same-sector peer-distance features to ψ recovered roughly one fifth of that gap (Table 2), narrowing the holdout generalization gap from +4.99% to +4.36% IV; the residual +4.36% portion arose from earnings surprises whose direction and magnitude could not be inferred from the calendar alone. We then exercised the calibrated framework as a synthetic-data generator on a real GS 2026-05-29 \$890 put and \$970 call (both 31 DTE, 30-delta) pulled from the 04-28-2026 capture, forward-simulated 1,000 price paths under the JumpHMM and the per-ticker ψ_{NN}, and tracked path-conditional implied volatility, premium decay, finite-difference

Leisen-Reimer Greeks, and terminal short-premium P&L from a single coherent simulation. The short put and short call kept their full premium on 72% and 64% of paths respectively, and the asymmetric tail risk of the short call (worst-case loss 1.50× that of the short put despite a smaller premium received; Supplementary Table S8, Fig. 5) emerged naturally from the negative skew that the per-ticker ψ_{NN} encoded together with the leverage coupling that lifted v_t on down-paths. A cross-ticker re-run on LLY (Fig. 6, Supplementary Table S9) reproduced the same qualitative asymmetry on a ticker with a different sector and a higher IV regime, confirming the result was structural rather than ticker-specific. The framework was implemented as an open-source Julia package and is available at https://github.com/varnerlab/heston-implied-volatility-model.

Code and Data Availability

The framework was implemented as an open-source Julia package, `HestonIV`, available at https://github.com/varnerlab/heston-implied-volatility-model. The repository ships the 15-date, 31-ticker, 6-sector option-ladder CSVs used for calibration; the pretrained JumpHMM portfolio surrogate; the calibrated ψ_{NN} cache (`calibrate_ladders_per_ticker_nn_cache.jld2`) that backs the forward scenarios; and per-scenario simulation caches (`{ticker}_short_premium_simulation_cache.jld2`) that reproduce the figures without re-running the JumpHMM forward pass. The earnings calendar used for the temporal-holdout features is included under `code/data/earnings/`. Setup, dependency-pin (`Project.toml`/`Manifest.toml`), and one-line invocations for every script that produces a paper figure are documented in `SETUP.md`. The walk-forward, static-arbitrage, and SABR-baseline scripts used in the revision are co-located under `code/examples/`.

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A Derivation of the per-contract variance decomposition $\sigma_t^2 = v_t$

The construction of the per-contract instantaneous variance in (8) is direct once the dependence of the mean-reversion target $\theta(i, t; K)$ in (4) on contract coordinates is recognized as the channel through which the smile and term-structure shape ψ enters the model. The shape factor enters the model exactly once, in the SDE drift, and the variance state v_t inherits both the level and the contract-level smile geometry from the drift through mean-reversion; no separate multiplicative shape factor on v_t is required to recover the per-contract IV. This appendix expands that construction, derives the consistency of the entry-time identity $\sigma_0^2 = \theta(i, 0; K)$ with the stationary law of the SDE, and shows how the same expression specializes to the observation-day calibration form (5).

The modified Heston SDE (3) reads

$$dv_t \;=\; \kappa\big(\theta(i,t;\,K) - v_t\big)\,dt \;+\; \sigma_v\sqrt{v_t}\,dW_t, \tag{16}$$

in which the mean-reversion target $\theta(i, t; K) = \theta_{i,s_t}(1 + \gamma M_t)\,\psi(\mathrm{DTE}_t, K/S_t, e_{i,t}, e_{i,t}^{\mathrm{peer}})$ from (4) depends on contract coordinates through ψ and on regime state and aggregate mood through the level prefactor. Because the drift carries the K and DTE dependencies, the variance process v_t is not a single scalar trajectory per ticker per path but a family of trajectories indexed by contract: each (ticker, K, DTE) triple obtains its own v_t under the same Brownian forcing W_t with a contract-specific drift target. The price path S_t enters the drift only through the moneyness K/S_t in ψ, so contracts sharing a path see correlated drift modulations whenever the underlying moves but each evolves its own variance state from those modulations. The per-contract instantaneous variance defined in (8) as $\sigma_t^2(K, S_t) = v_t$ then inherits the smile and term structure directly from ψ via the drift; the shape factor ψ is not reapplied as a multiplicative overlay on v_t, since the SDE has already absorbed it into the variance dynamics.

At the contract entry time $t = 0$ the construction reproduces the calibrated surface exactly rather than in expectation. The initial condition $v_0 = \theta(i, 0; K)$ from (7) gives $\sigma_0^2(K, S_0) = v_0 = \theta_{i,s_0}(1 + \gamma M_0)\,\psi(\mathrm{DTE}_0, K/S_0, e_{i,0}, e_{i,0}^{\mathrm{peer}})$ at the contract's entry coordinates; under the calibration restrictions $\gamma = 0$ and $\theta_{i,s_0} = \theta_i$ this collapses to $\theta_i \cdot \psi(\cdot) = \theta_{\mathrm{cal}}(i, t)$ from (5), which is the calibrated decomposition (11). The calibrated ψ surface from the multi-date ladder corpus therefore also serves as the entry IV surface at simulation start without requiring a separate v_0 degree of freedom or an external IV input. Beyond the entry boundary the same identity holds in expectation under the stationary law of (16): conditioning on the slow variables (s_t, M_t) and the contract coordinates $(K, \mathrm{DTE}_t, S_t, e_{i,t}, e_{i,t}^{\mathrm{peer}})$ under which the drift target is constant, v_t obeys a CIR process whose stationary mean is the target itself, so $\mathbb{E}[v_t \mid \cdot] \to \theta(i, t; K)$ and $\mathbb{E}[\sigma_t^2 \mid \cdot] \to \theta(i, t; K)$ as $\kappa t \to \infty$. The conditional variance of the stationary law is $\sigma_v^2\theta(i, t; K)/(2\kappa)$, so the fluctuations of σ_t^2 around the target are controlled jointly by vol-of-vol and the mean-reversion rate, and the reflecting boundary at $v = 0$ in the discretization (10) prevents excursions to negative variance under non-Feller parameter combinations.

The calibration form (5) arises by specializing this construction to a single observation day.

There is no time evolution within an observation-day IV surface, so v_t is taken at its entry value $\theta(i,0;K)$ over the day, with regime and mood absorbed into the per-ticker level θ_i, and the calibration target reduces to $\sigma^2_{\text{cal}}(K,S) = \theta_i \cdot \psi(\cdot) = \theta_{\text{cal}}(i,t)$. The calibration loss in (13) fits $\{\theta_i\}_i$ and ψ jointly to market IV under this stationary form, with no SDE solve required. The three symbols carry distinct roles throughout: $\theta(i,t;K)$ is the mean-reversion target of the per-contract variance, evaluated at the contract's coordinates at each time step; v_t is the variance state itself, evolving per contract under (16) with that target; and ψ is the contract-coordinate-dependent shape factor that enters the model once, through the drift target, and is carried through to the per-contract IV $\sigma_t^2 = v_t$ by the variance dynamics rather than by a separate multiplication.

B Sector-NN calibration: corpus partition and per-sector breakdown

We decomposed the headline three-way RMSE comparison along two axes and observed that the ETF and Technology sectors together carried roughly three-quarters of the corpus, the sector NN reduced RMSE in every sector over the shared NN with the largest gains in ETF and financials, and the per-ticker residual error concentrated on a small set of technology and healthcare names (Tables S1, S2, S3). The corpus partition that defined the sector grouping is reported in Table S1, the per-sector RMSE decomposition in Table S2, and the per-ticker fit ranking under the sector NN in Table S3.

Table S1: Sector partition of the 31-ticker ladder. Each ticker appeared on at least one of fifteen capture dates (2026-04-14 through 2026-05-11); per-sector observation counts were pooled across dates. ETF observation counts were substantially larger than other sectors because SPY, QQQ, and IWM carried far deeper strike ladders than individual-name chains.

Sector	Observations	Tickers
ETF	102,738	IWM, QQQ, SPY
Technology	68,334	AAPL, AMD, AVGO, GOOG, INTC, META, MSFT, MU, NVDA, QCOM
Healthcare	26,387	ABBV, AMGN, BMY, JNJ, LLY, MRNA, PFE, UNH
Financials	17,938	BAC, GS, JPM, WFC
Retail	11,320	TGT, UPS, WMT
Energy	7,832	CVX, OXY, XOM

B.1 Leave-one-date-out validation on the fifteen-date corpus

We held out the latest capture (2026-05-11) and retrained the sector ψ_{NN} on the remaining fourteen dates as a leave-one-date-out check, and observed that the held-out test RMSE landed within 0.68% IV of the pooled train RMSE (Table S4). The held-out day comprised 14,442 observations (6.2% of the corpus); we trained the six sector networks on the remaining fourteen dates using identical architecture, schedule, and seed, standardised only on the training rows so the test split could not

Table S2: Per-sector RMSE (% IV) across the three calibration strategies on the 31-ticker ladder. The sector NN reduced RMSE in every sector over the shared NN, with the largest absolute gains in ETF and financials, where the global shape poorly represented the local geometry. Healthcare and technology retained the largest residual errors, reflecting within-sector smile heterogeneity that a single shared sector ψ could not absorb.

Sector	Parametric	Shared NN	Sector NN	Obs. count
ETF	10.54	8.69	6.10	102,738
Financials	9.05	7.79	6.70	17,938
Energy	9.36	8.24	7.88	7,832
Retail	11.73	10.23	9.50	11,320
Healthcare	13.49	12.71	12.53	26,387
Technology	15.58	15.31	14.47	68,334

leak through the input scaling, and evaluated test RMSE on the held-out day. The overall train RMSE of 10.21% matched the full-corpus in-sample number of 10.24% within rounding, confirming that withholding one day did not perturb the fit, and the held-out test RMSE of 10.89% left a generalization gap of only +0.68% IV pooled across all six sectors. The gap was strongly sector-specific: four of six sectors (ETF, Financials, Healthcare, Tech) carried negative or small-positive gaps, while Energy (+2.23%) and Retail (+3.96%) showed larger positive gaps consistent with the small per-sector sample sizes on the holdout day amplifying single-name idiosyncrasies. The overall pattern was consistent with the temporal-holdout result: with no earnings prints inside the holdout, the sector NN generalized to an unseen capture date with $\sim 7\%$ relative degradation in RMSE, far below what an overfit model would exhibit.

B.2 Walk-forward temporal validation across the fifteen-date corpus

We extended the leave-one-date-out check to a full walk-forward sweep across the fifteen-date corpus: for $k = 5, \ldots, 14$, we trained the sector ψ_{NN} on the first k sorted ladder dates (Configuration-A pooled control, 2-input ψ) and tested on date $k+1$, producing ten folds (Table S5). The fold-wise pattern was the same one isolated by the earnings-holdout study: median test RMSE was 11.08% IV and median generalization gap was +2.02%, but the distribution was driven by a small number of earnings-blowup folds rather than uniform temporal degradation. The largest single-fold gap was +9.41% on the $k = 6$ fold (test day 2026-04-23), the same INTC print day that drove the headline Configuration-A gap of +4.99% in Table 2; that fold tested on 04-23 alone, while Configuration-A averaged 04-23 with the calmer 04-24, which is why the standalone fold gap was roughly twice as large. The non-earnings $k = 14$ fold (test day 2026-05-11, no earnings prints in the window) produced a gap of +0.68%, reproducing the leave-one-date-out result of Table S4 and confirming that the multi-date pool generalizes faithfully when no scheduled event sits inside the test window.

Table S3: Best and worst individual ticker fits under the sector-specific NN calibration, ranked by RMSE across all 31 tickers. The best fits were dominated by the broad-market ETFs and the deep-liquidity financials (GS, JPM); the worst fits were concentrated in technology (QCOM, MU, INTC, AMD) and healthcare (PFE, MRNA), where within-sector smile heterogeneity was largest. PFE and BMY remained the only tickers below the 2,000-observation per-ticker threshold.

Rank	Ticker	Obs. count	RMSE (% IV)	Sector
Best 5				
1	GS	8,772	5.40	Financials
2	SPY	42,224	6.02	ETF
3	IWM	21,368	6.11	ETF
4	QQQ	39,146	6.17	ETF
5	JPM	3,720	6.18	Financials
Worst 5				
27	PFE	1,300	17.66	Healthcare
28	AMD	4,566	17.74	Technology
29	INTC	3,417	17.78	Technology
30	MU	6,619	18.45	Technology
31	QCOM	3,375	21.46	Technology

B.3 Static-arbitrage diagnostic on the calibrated ψ surfaces

We tested the calibrated ψ surfaces against the standard no-arbitrage conditions $\partial^2 C/\partial K^2 \geq 0$ (butterfly) and $\partial C/\partial T \geq 0$ (calendar) on a 20×20 moneyness–maturity grid ($K/S \in [0.80, 1.20]$ log-spaced, DTE $\in [7, 365]$ log-spaced), pricing European calls through the same 200-step CRR pricer used elsewhere in the calibration and evaluating central second differences in K and forward first differences in T on each grid. Across the six sector surfaces (one representative ticker per sector, with θ_i taken from that ticker) and the two per-ticker surfaces used in the forward scenarios (GS and LLY), butterfly violations averaged 9.76% of grid points and calendar violations averaged 0.76% (Table S6). Calendar violations were essentially absent on every surface, consistent with the monotone term-structure shape the network learned; butterfly violations clustered in the deep moneyness wings ($K/S \lesssim 0.85$ and $K/S \gtrsim 1.15$) where the corpus carried fewer observations and the network had to extrapolate the curvature. The unconstrained neural surrogate has no structural guarantee of butterfly convexity, and we did not impose a no-arbitrage penalty during training; we report the violation rates here as a known limitation rather than a corrected feature, with no-arbitrage-aware extensions reserved for future work.

Table S4: Leave-one-date-out validation of the sector NN, holding out 2026-05-11. Per-sector train and test RMSE (% IV) and gap (test − train). The pooled gap of +0.68% IV is small relative to the in-sample RMSE itself, indicating that the architecture does not overfit the fifteen-date corpus.

Sector	Train (%)	Test (%)	Gap (%)
ETF	6.12	5.87	−0.25
Financials	6.72	6.32	−0.40
Healthcare	12.58	12.00	−0.58
Tech	14.45	15.04	+0.58
Energy	7.74	9.97	+2.23
Retail	9.21	13.17	+3.96
Pooled	10.21	10.89	+0.68

Table S5: Walk-forward temporal validation across the fifteen-date corpus. For each k, the sector ψ_{NN} was trained on the first k sorted ladder dates and tested on date $k+1$; standardization was computed from the training rows only. The earnings column flags folds in which any test-day row sat within ± 3 days of an earnings print for the ticker or any same-sector peer. The $k = 6$ fold (test 2026-04-23) reproduces the INTC-print blowup that drives the Configuration-A gap in Table 2; the non-event $k = 14$ fold reproduces the leave-one-date-out result of Table S4.

k	Test date	N_{tr}	N_{te}	Train (%)	Test (%)	Gap (%)
5	2026-04-22	88,346	15,551	7.61	10.47	+2.87
6	2026-04-23	103,897	11,482	8.05	17.45	+9.41
7	2026-04-24	115,379	15,351	9.34	7.79	−1.55
8	2026-04-27	130,730	11,379	9.16	12.57	+3.41
9	2026-04-28	142,109	16,776	9.42	12.86	+3.44
10	2026-04-29	158,885	16,811	9.78	13.12	+3.34
11	2026-05-01	175,696	14,825	10.11	10.08	−0.03
12	2026-05-06	190,521	17,293	10.10	11.28	+1.18
13	2026-05-08	207,814	12,293	10.18	10.82	+0.64
14	2026-05-11	220,107	14,442	10.21	10.89	+0.68
Median across 10 folds				—	11.08	+2.02
Mean across 10 folds				—	11.73	+2.34

Table S6: Static no-arbitrage diagnostic on eight calibrated ψ surfaces, evaluated on a 20×20 moneyness–maturity grid via 200-step CRR European call prices. Butterfly violations are grid points where $\partial^2 C/\partial K^2 < 0$ (computed by central second difference on the 18 inner strikes per row, 360 cells per surface); calendar violations are grid points where $\partial C/\partial T < 0$ (forward first difference, 380 cells per surface).

Surface	Butterfly viol.	Frac. (%)	Calendar viol.	Frac. (%)
ETF / SPY (sector)	56 / 360	15.56	23 / 380	6.05
Tech / NVDA (sector)	17 / 360	4.72	0 / 380	0.00
Healthcare / LLY (sector)	20 / 360	5.56	0 / 380	0.00
Financials / GS (sector)	54 / 360	15.00	0 / 380	0.00
Retail / WMT (sector)	45 / 360	12.50	0 / 380	0.00
Energy / XOM (sector)	24 / 360	6.67	0 / 380	0.00
GS per-ticker	36 / 360	10.00	0 / 380	0.00
LLY per-ticker	29 / 360	8.06	0 / 380	0.00
Aggregate	281 / 2,880	9.76	23 / 3,040	0.76

C Holdout diagnostics motivating the earnings-aware mechanism

We decomposed the holdout test error on the Tech sector across the pooled control (A) and the earnings-aware fit (C) and traced the Configuration-C improvement to the peer-coupling feature: tickers whose own prints fell outside the holdout but whose sector peers printed during it received the largest test-RMSE reductions, while INTC, the print-day ticker, was the one Tech name to worsen (Table S7). The decomposition rested on two preceding diagnostics that established the holdout RMSE jump as a localized earnings response rather than a broad-market regime shift. First, the broad-market volatility index VIX traced a flat path through the holdout window: VIX closed at comparable levels on the final training day and the two holdout days, with no intraday excursion during the holdout that approached even a one-standard-deviation move relative to the trailing six-day window. A regime-shift hypothesis required the index to move materially over the holdout, and it did not. Second, the per-ticker ATM IV trace revealed that the IV expansion on 04-23 and 04-24 was concentrated entirely in the technology sector, and within that sector concentrated on the names with earnings prints inside or adjacent to the holdout window: INTC (which printed inside the window), MSFT, META, and GOOG all exhibited ATM IV elevations of 5 to 30 percentage points relative to their training-block baselines, while non-Tech names traced ATM IV paths consistent with their pre-holdout levels. The combination established that the holdout RMSE jump was not a market-wide effect that the architecture had failed to absorb but a localized response to the earnings calendar, and it directly motivated the earnings indicator and peer-coupling feature reported in Configuration C of Table 2. INTC, the print-day ticker, was the one Tech name to worsen under the earnings-aware fit because its +2192% EPS surprise produced a post-print IV crush that no calendar-based indicator could anticipate.

Table S7: Per-ticker test RMSE on the Tech sector across the pooled control (A) and the earnings-aware fit (C). Columns e and e^{peer} report the median signed days-to-earnings and the minimum same-sector peer distance for each ticker on the holdout days. The largest gains accrued to tickers whose peers printed within the holdout (QCOM, AMD, META, MSFT, GOOG); INTC, the print-day ticker, was the one name that worsened, reflecting an unanticipated outcome that no calendar-based indicator could absorb.

Ticker	e	e^{peer}	A: pooled (%)	C: earnings-aware (%)	Δ (%)
QCOM	+5	1	41.66	37.59	−4.07
AMD	+11	1	40.07	36.50	−3.57
META	+5	1	14.32	10.73	−3.59
MSFT	+5	1	14.00	10.73	−3.28
GOOG	+5	1	9.87	7.90	−1.97
MU	−30	1	16.95	15.59	−1.36
AAPL	+6	1	10.42	10.34	−0.08
AVGO	+30	1	12.58	12.83	+0.24
NVDA	+26	1	9.43	10.47	+1.05
INTC	−1	5	33.17	35.94	+2.77

D Short-premium scenarios: P&L statistics and cross-ticker validation

We tabulated the moments and tail statistics per contract for the GS 31-day forward simulation and observed the short-premium asymmetry that the main-text figures rendered qualitatively: a thin-tailed right side with a hard ceiling at the entry premium, paired with a long left tail that extended an order of magnitude beyond the median (Table S8). The model's $t = 0$ Leisen-Reimer fair value sat essentially at the market mid on both contracts (entry edge +\$0.75 on the put, −\$0.02 on the call), confirming the per-ticker ψ_{NN} matched the 04-28 GS market IV. The dispersion was larger on the call leg (std \$51.58 vs. \$34.12), and the worst-case loss on the call (−\$484.57) exceeded the worst put loss (−\$323.42) by a factor of 1.50, despite the call collecting a smaller entry premium.

Table S8: Terminal P&L statistics from 1,000 GS 31-day forward simulations of the real 2026-05-29 \$890 put and \$970 call. Entry premium was the market mid from the 2026-04-28 capture (GS spot $S_0 = \$926.38$). The model's $t = 0$ Leisen-Reimer fair value at NN-IV was reported as a reference; statistics are per contract.

Statistic	Short put (\$)	Short call (\$)
Strike K	890	970
Market mid (entry premium)	+16.51	+16.09
Model $t = 0$ fair value	+17.26	+16.07
Entry edge (model − market)	+0.75	−0.02
Mean P&L	+1.52	−9.00
Median P&L	+16.51	+16.09
Std P&L	34.12	51.58
5%-tile P&L	−76.13	−113.16
Worst-case P&L	−323.42	−484.57
Premium kept in full	72.2%	64.3%

We re-ran the GS pipeline verbatim on an LLY option pair from the same 04-28 capture and observed that the qualitative features (tall premium-ceiling spike, long left tail, fatter short-call wing, leverage-driven Vega expansion on down-paths) reproduced on a ticker with a different sector, IV regime, and JumpHMM marginal (Table S9; Figs. S2, S3, S4). Terminal P&L statistics per contract are reported in Table S9; the share-price-and-premium panel (Fig. S2), IV trajectories (Fig. S3), and Leisen-Reimer American Greek panels (Fig. S4) parallel the GS figures in the main text.

D.1 Cross-sector short-premium scenarios on five additional tickers

We re-ran the short-premium scenario pipeline verbatim on five additional tickers spanning ETF, Energy, Tech, low-liquidity Healthcare (PFE, sector-NN fallback), and large-cap Healthcare (Table S10). The short-premium asymmetry reproduced cleanly on AMD (Tech, $\sim 65\%$ IV regime), where the worst-call loss exceeded the worst-put loss by 3.2×, and collapsed to near-symmetry on

Algorithm 2 Path-conditional Greeks via central finite differences on the LR pricer

Require: State (S, σ, τ) with $\tau = T - t$; contract (K, r); LR depth N_{LR}; spot bump h_S (relative); IV bump h_σ (absolute).

Ensure: Δ, Γ, Vega.

$P_+^S \leftarrow \text{LR}_{N_{\text{LR}}}(S(1+h_S),\, K,\, \sigma,\, \tau,\, r)$

$P_-^S \leftarrow \text{LR}_{N_{\text{LR}}}(S(1-h_S),\, K,\, \sigma,\, \tau,\, r)$

$P_0 \leftarrow \text{LR}_{N_{\text{LR}}}(S,\, K,\, \sigma,\, \tau,\, r)$

$P_+^\sigma \leftarrow \text{LR}_{N_{\text{LR}}}(S,\, K,\, \sigma + h_\sigma,\, \tau,\, r)$

$P_-^\sigma \leftarrow \text{LR}_{N_{\text{LR}}}(S,\, K,\, \sigma - h_\sigma,\, \tau,\, r)$

$\Delta \leftarrow \dfrac{P_+^S - P_-^S}{2\, h_S\, S}$

$\Gamma \leftarrow \dfrac{P_+^S - 2\, P_0 + P_-^S}{(h_S\, S)^2}$

$\text{Vega} \leftarrow \dfrac{P_+^\sigma - P_-^\sigma}{2\, h_\sigma}$ $\triangleright$ per unit IV; multiply by 0.01 for per-1% IV

return $(\Delta, \Gamma, \text{Vega})$

SPY (ETF, balanced moneyness profile and shallow smile). The PFE row exercised the sector-NN fallback path of the framework (PFE has 1,300 ladder observations, below the 2,000-observation per-ticker threshold, so its ψ_{NN} is the Healthcare sector network rather than a per-ticker fit) and produced an operationally clean short-premium distribution despite the small absolute dollar values that the $\sim$ \$26 underlying generated. JNJ reproduced the LLY-style negative entry edge on its put leg ($-$\$0.98 at model fair value below market mid), reflecting the same time-averaged-ψ vs. capture-day-regime gap noted for LLY on 04-28.

Table S9: Terminal P&L statistics from 1,000 LLY 31-day forward simulations of the real 2026-05-29 \$825 put and \$940 call. Entry premium was the market mid from the 2026-04-28 capture (LLY spot $S_0 = \$873.83$). LLY had dropped $\sim 5\%$ on 04-28, so the contracts were captured against an elevated-IV regime relative to the fifteen-date calibration window; the model's $t = 0$ Leisen-Reimer fair value at the per-ticker ψ_{NN}-implied IV came in below the market mid on both legs, with the entry edges reflecting the gap.

Statistic	Short put (\$)	Short call (\$)
Strike K	825	940
Market mid (entry premium)	+23.30	+20.76
Model $t = 0$ fair value	+21.95	+18.96
Entry edge (model − market)	−1.35	−1.80
Mean P&L	+14.31	+6.45
Median P&L	+23.30	+20.76
Std P&L	23.75	36.81
5%-tile P&L	−41.54	−79.68
Worst-case P&L	−185.19	−303.88
Premium kept in full	77.4%	75.7%

Table S10: Short-premium scenarios on five additional tickers spanning ETF, Energy, Tech, low-liquidity Healthcare, and large-cap Healthcare. Each scenario was a 31-day forward simulation of 1,000 paths from the ticker's pretrained JumpHMM marginal, with 30Δ put and call strikes drawn from the 2026-04-28 capture and contracts expiring 2026-05-29. All five scenarios used the same Heston backbone $(\kappa, \sigma_v, \rho) = (15.0, 0.5, -0.6)$, the same 201-step Leisen-Reimer American pricer, and identical caching/seeding as the GS and LLY scenarios; only the per-ticker ψ_{NN} surface, the long-run-drift anchor, and the contract specifications varied across rows. PFE's ψ source is the Healthcare sector network because PFE has 1,300 ladder observations, below the per-ticker fit threshold.

Ticker	Sector	ψ source	K_p	K_c	Edge$_p$ (\$)	Edge$_c$ (\$)	Worst$_p$ (\$)	Worst$_c$ (\$)	Kept$_p$ (%)	Kept$_c$ (%)
SPY	ETF	per-ticker	696	729	−0.07	+0.94	−132.65	−132.59	73.8	59.3
XOM	Energy	per-ticker	145	160	−0.46	+0.32	−36.19	−56.40	70.4	72.1
AMD	Tech	per-ticker	300	365	+0.15	−0.31	−111.84	−361.75	68.9	71.8
PFE	Healthcare	sector	26	28	+0.18	+0.30	−6.21	−5.70	64.8	73.7
JNJ	Healthcare	per-ticker	220	235	−0.98	+0.45	−40.32	−36.42	78.5	64.4

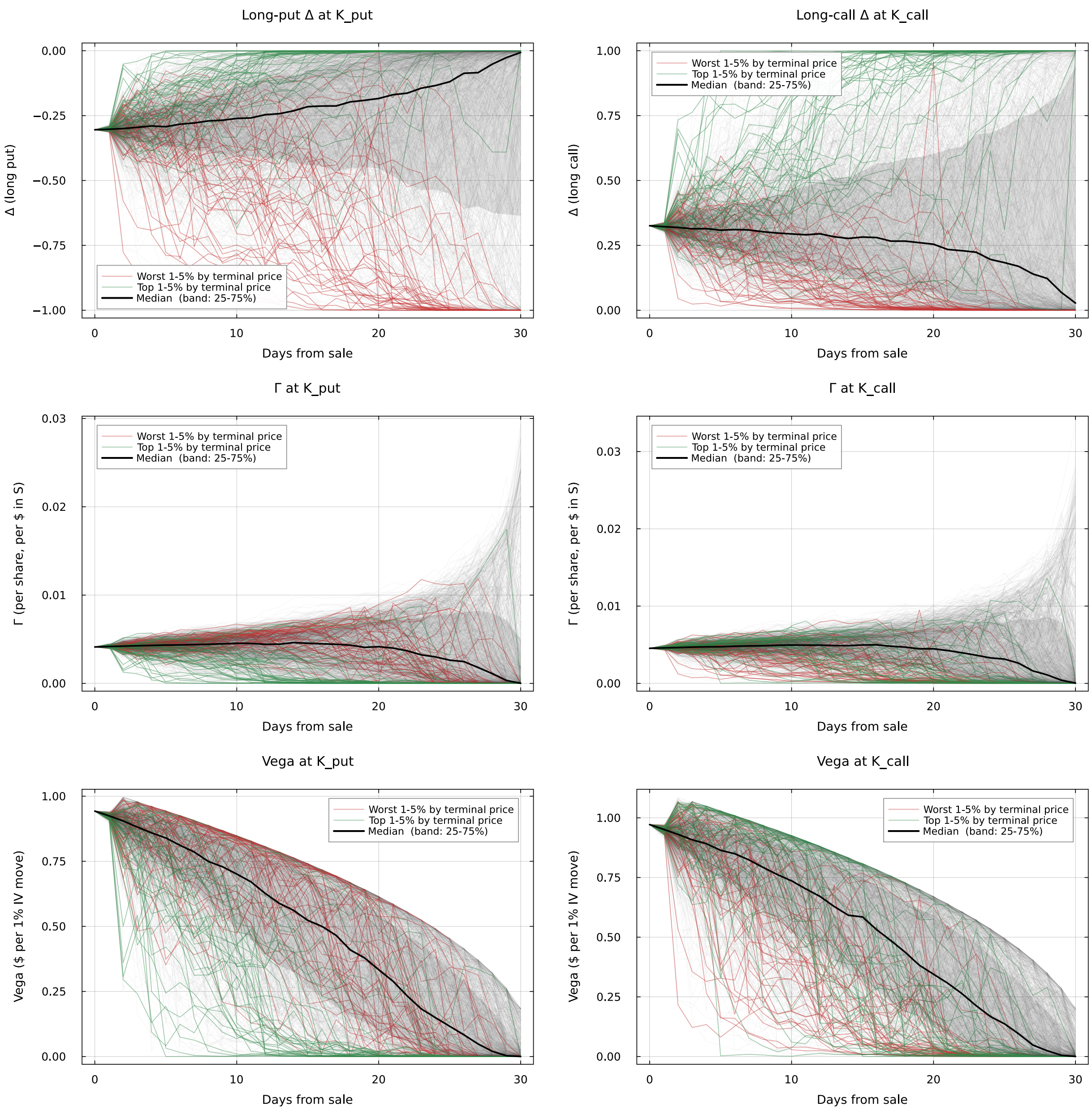

Figure S1: Leisen-Reimer American Greeks (Δ, Γ, Vega) at the GS put and call strikes over the 31-day horizon, computed by central finite differences (Algorithm 2) on the 201-step Leisen-Reimer pricer at the per-path NN-IV ($h_S = 1.5\% S_t$ for Δ/Γ, $d\sigma = \pm 0.005$ for Vega). Long-Greek convention: short-position Greeks are the negatives. Top row: Δ. Middle: Γ, exhibiting the short-gamma spike on paths that pinned to the strike near expiry. Bottom: Vega, decaying toward zero with $\sqrt{T-t}$ on every path. Tail overlays as in Fig. 3.

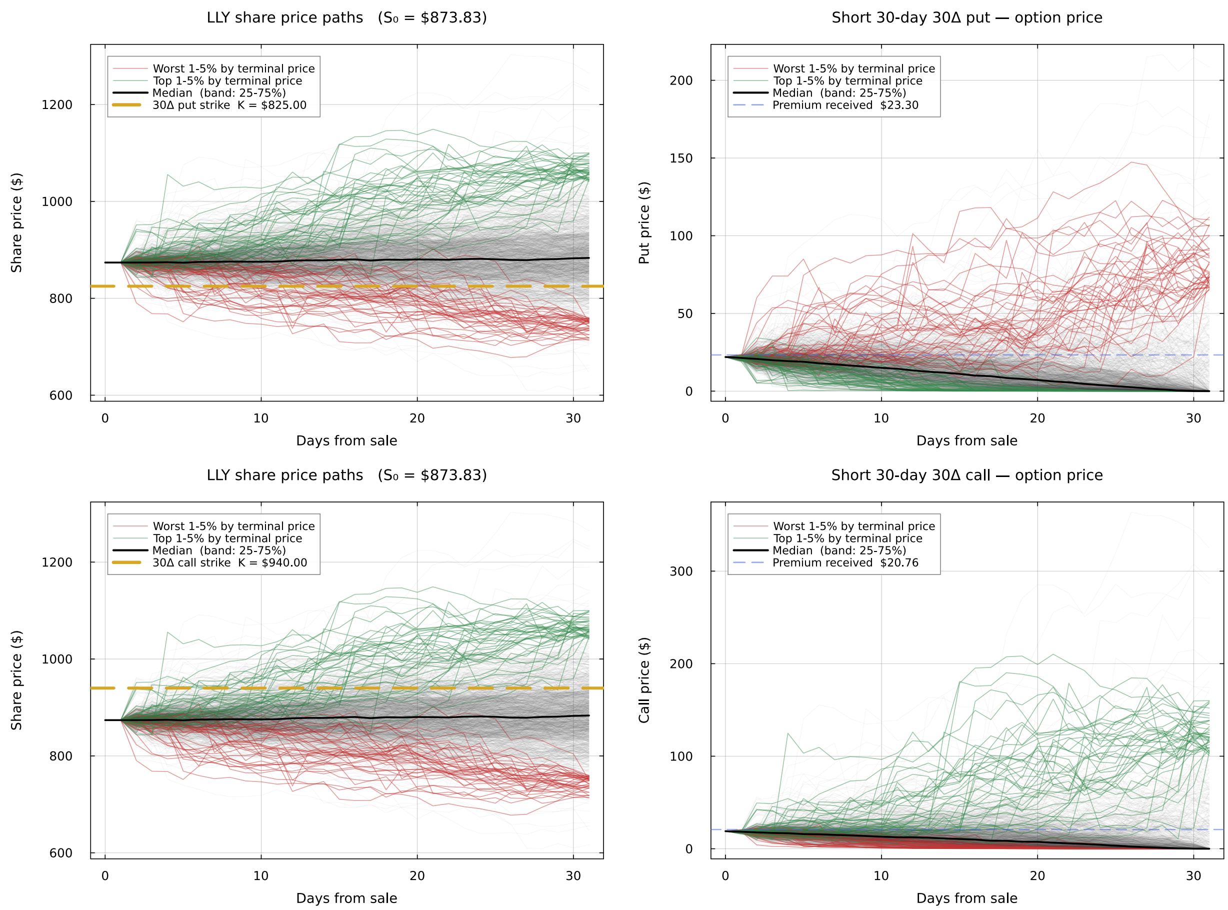

Figure S2: Forward LLY share-price paths and short option premium paths for the real LLY 2026-05-29 contracts, 1,000 simulated trajectories over 31 days; analogue of GS Fig. 3. Top row: share path with $K_p = \$825$ marked (left) and short-put premium (right), with the worst 1–5% by terminal price overlaid in red. Bottom row: share path with $K_c = \$940$ marked (left) and short-call premium (right), with the top 1–5% overlaid in green. The $K_p = \$825$ strike sat $\sim 6\%$ below spot, further from ATM than the GS put because LLY's higher IV regime placed the same-delta strike further from spot. Black solid: median; shaded band: 25–75% IQR. Blue dashed lines on the option-price panels mark the market mids received at sale ($23.30 put, $20.76 call).

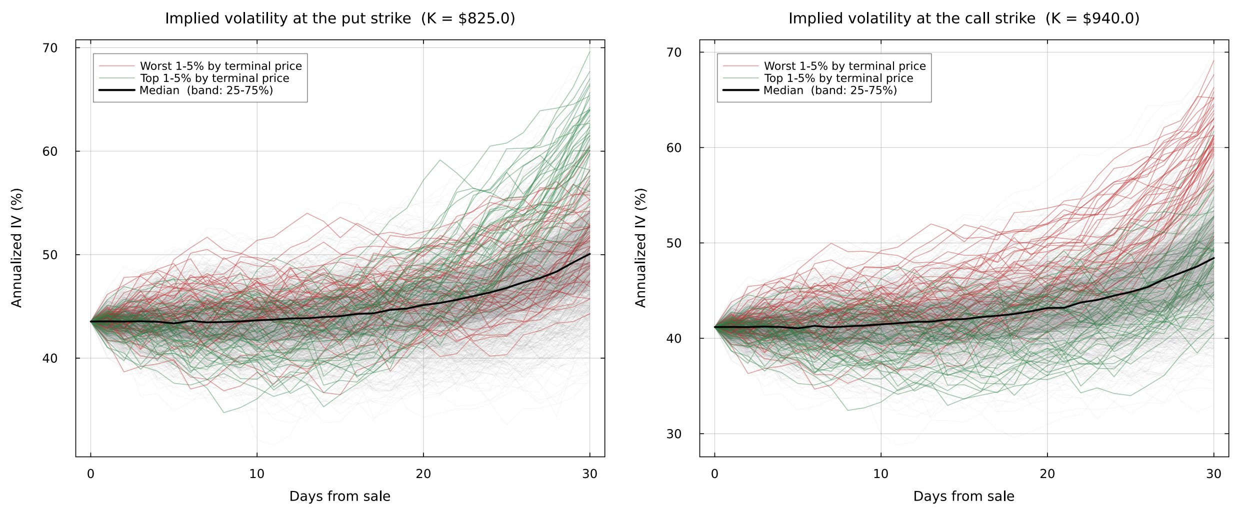

Figure S3: Path-conditional IV at K_{put} and K_{call} over the 31-day horizon; analogue of GS Fig. 4. IV expanded on worst-5% down-paths and contracted on top-5% up-paths, consistent with the negative leverage coupling $\rho = -0.6$ that lifted v_t when shares fell.

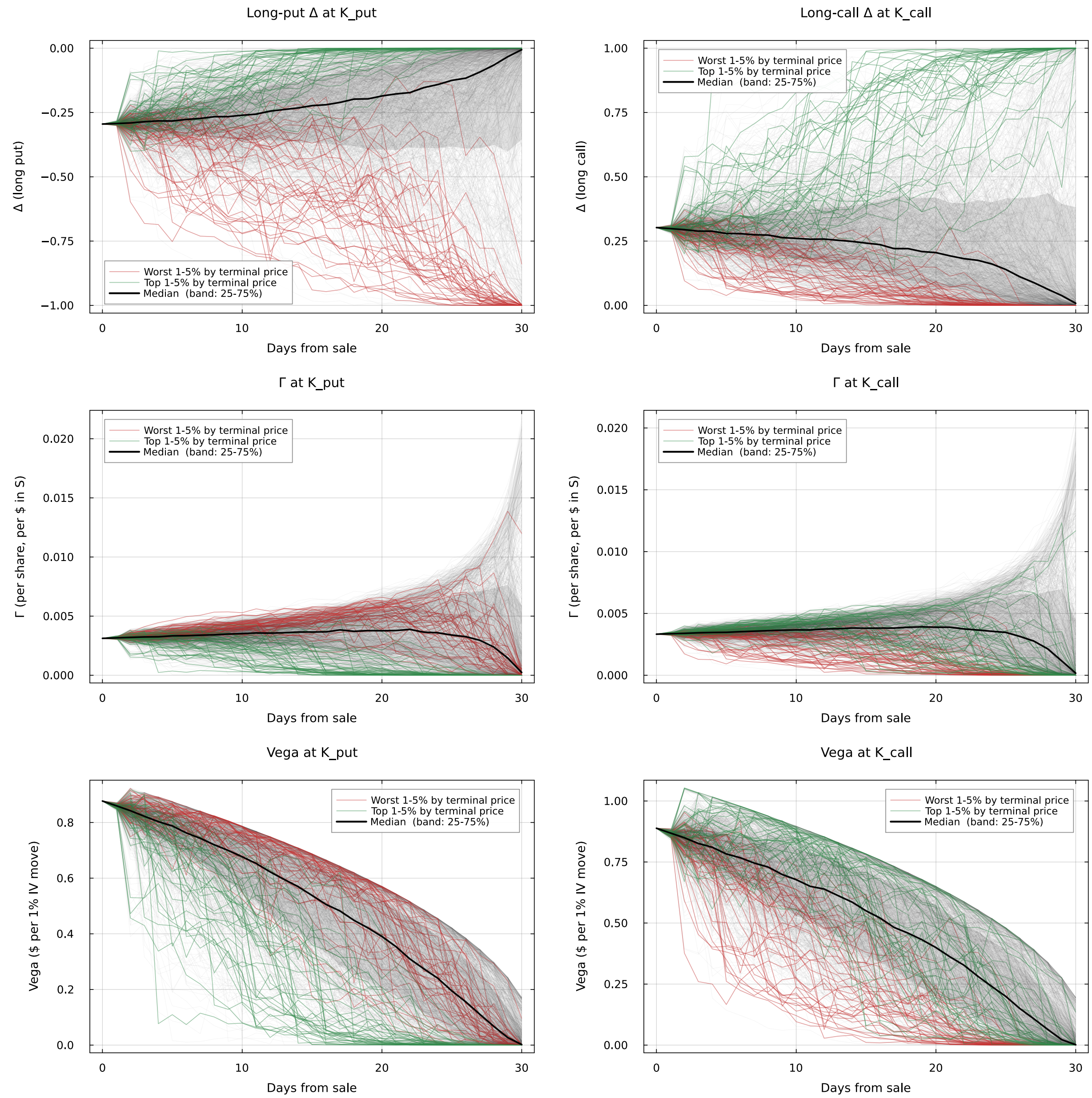

Figure S4: Leisen-Reimer American Greeks (Δ, Γ, Vega) at the LLY put and call strikes; analogue of GS Fig. S1. Short-position Greeks are the negatives of the displayed long-side values. The short-gamma spike near expiry and the $\sqrt{T-t}$ Vega decay both reproduced the patterns seen on GS.

E Adaptive recalibration via low-rank daily adapters: empirical detail

This appendix tabulates the empirical detail behind the adaptive recalibration: the 20-cell sweep, the no-arbitrage audit on the V1-adapted surfaces, the trigger ROC sweep, and the LLY 04-28 case study.

E.1 20-cell sweep

Cells were selected deterministically from the 31×15 corpus after the per-cell base-RMSE scan: the five highest-RMSE cells (problem days, 50.6%–68.6% base RMSE), ten cells stratified through the middle of the RMSE distribution (6.8%–9.5%), and the five lowest-RMSE cells (3.4%–4.1%, all GS captures where the per-ticker $\psi_{\rm NN}$ was already a tight fit). V0 trained only the scalar $\Delta \log \theta_i$ on the day's slice; V1 trained the rank-2 LoRA across all three Dense layers plus $\Delta \log \theta_i$ (135 trainable parameters per cell). Both variants used Adam with patience-150 early stopping; convergence was reached within 300–600 epochs on every cell.

E.2 LLY 04-28 entry-edge case study

The LLY 04-28 cell appeared as a single row in Table S11 (sweep position: middle cell, base RMSE 15.66%, V1 RMSE 4.34% reported here in detail rather than the 04-22 row above), and we re-priced its 30Δ put and call under each adapter variant to translate the slice-RMSE improvement into a dollar entry-edge story.

E.3 No-arbitrage audit on V1-adapted surfaces

The rank-2 adapter carries no structural arbitrage constraint, so we re-evaluated the static-arbitrage diagnostic on each of the 20 cells under the V1-adapted $\psi_{\rm NN}$ and compared to the base. The aggregate butterfly violation rate rose from 9.78% (704/7,200) to 12.74% (917/7,200), and the calendar rate rose from 1.01% (77/7,600) to 2.32% (176/7,600). The amplification concentrated on the highest-base-RMSE cells where the adapter perturbation was largest in absolute terms (INTC 04-23, AMD 04-23, QCOM 04-23 each saw their butterfly rate triple), and six of the twenty cells actually saw violations decrease under adaptation. The unconstrained adapter is therefore not free: aggressive cells trade slice-RMSE accuracy for wing-arbitrage cleanliness. The natural extension — a Variant 2 with an arbitrage-penalty term in the adapter loss — is straightforward in the LoRA framework but was not implemented here.

E.4 Streaming trigger ROC

We evaluated the streaming refit trigger on every (ticker, date) cell in the 31×15 corpus (457 filtered cells after the $N_{\rm obs} \geq 10$ filter). Ground truth was defined as the binary indicator $\mathrm{RMSE}_t > 12\%$ IV (112/457 positive cells). The trigger was run with the absolute floor $\tau_{\rm abs} = 10\%$ fixed and

Table S11: Multi-cell LoRA sweep. Per-cell base, V0 (level-only), and V1 (rank-2 LoRA + level) RMSE on the day's ladder slice, with V1 adapter-norm summaries on the three Dense layers. V1 reduced RMSE on every cell (no overfit) and beat V0 across the board; improvements scaled with base RMSE. The five GS control cells (bottom block) saw V1 perturbations near zero, the desired behaviour on days the base already fit well.

Ticker	Date	N_{obs}	Base (%)	V0 (%)	V1 (%)	$\Delta \log \theta_{\text{V1}}$	$\lVert B_1 A_1 \rVert_F$	$\lVert B_2 A_2 \rVert_F$	$\lVert B_3 A_3 \rVert_F$
Problem cells (5 highest base RMSE)									
INTC	2026-04-23	421	68.58	17.07	5.97	+0.30	0.70	2.02	0.68
AMD	2026-04-23	510	55.57	46.84	41.06	+0.11	2.68	2.54	0.65
QCOM	2026-04-23	369	53.97	39.81	33.58	+0.13	1.40	1.64	0.42
AAPL	2026-04-29	533	40.90	35.30	30.58	+0.05	1.35	1.62	0.95
QCOM	2026-05-11	442	32.76	12.48	6.08	+0.14	0.80	1.19	0.36
Middle cells (10 stratified through the middle 50% of RMSE)									
META	2026-05-06	624	11.96	7.54	5.78	−0.06	0.78	0.69	0.24
AAPL	2026-04-23	521	10.97	10.44	9.08	−0.05	0.59	1.73	0.28
ABBV	2026-04-27	188	10.10	8.78	5.24	+0.09	0.70	0.71	0.46
IWM	2026-04-21	1,918	9.53	8.36	7.10	+0.05	0.53	0.99	0.12
MSFT	2026-05-01	624	8.98	6.48	5.33	−0.06	0.66	0.62	0.24
LLY	2026-04-22	598	8.54	8.16	4.59	+0.03	0.67	1.40	0.65
XOM	2026-04-29	188	7.97	7.26	5.85	+0.18	0.46	0.92	0.67
WFC	2026-04-27	144	7.50	7.44	5.72	−0.03	0.75	1.57	0.29
SPY	2026-04-21	2,701	7.14	6.89	5.61	+0.08	0.65	2.20	0.14
JNJ	2026-05-11	128	6.78	6.00	4.40	+0.01	0.30	1.38	0.29
Control cells (5 lowest base RMSE; near-zero V1 perturbation)									
GS	2026-04-22	583	4.13	4.01	3.15	−0.01	1.10	2.15	0.74
GS	2026-04-23	447	3.79	3.66	3.08	+0.04	0.47	1.06	0.55
GS	2026-04-16	806	3.70	3.70	3.22	+0.07	0.61	2.70	1.51
GS	2026-04-17	696	3.57	3.36	2.95	+0.02	1.08	1.35	0.29
GS	2026-04-24	517	3.41	3.40	2.73	+0.05	0.82	1.11	0.79
Median problem-cell improvement (% IV)				—	—	+20.4			
Median middle-cell improvement (% IV)				—	—	+2.4			
Median control-cell improvement (% IV)				—	—	+0.7			

the relative threshold τ_z swept from 0 to 6. The trailing window length was 4 non-flagged dates per ticker, with $1.4826 \times \text{MAD}$ as the robust scale. Flagged dates were excluded from the trailing baseline for subsequent decisions on the same ticker.

Table S12: LLY 04-28 entry-edge case study, the failure case that motivated the adapter. Market mids and ψ_{NN}-implied fair values are reported alongside V0 (level-only) and V1 (rank-2 LoRA + level). Variant 0 over-prices both 30Δ contracts because the global level lift needed to fit the wings over-shoots the 30Δ strikes; Variant 1's rank-2 shape perturbation lifts the wings while leaving the near-the-money region near market, closing the −\$1.80 call edge to +\$0.46 and the −\$1.35 put edge to −\$1.15.

	Market mid	Base FV	V0 FV	V1 FV	Base edge	V0 edge	V1 edge
LLY \$825 put	\$23.30	\$21.95	\$29.45	\$22.15	−\$1.35	+\$6.15	−\$1.15
LLY \$940 call	\$20.76	\$18.96	\$26.20	\$21.22	−\$1.80	+\$5.44	+\$0.46
Slice RMSE on 04-28 (% IV)	—	15.66	11.78	**4.34**	—	—	—

Table S13: Streaming-trigger operating-point summary on the 31×15 corpus (457 cells, 112 truth-positive). The Youden-optimal point ($\tau_z = 0$) recovers 74% of refit-needed cells with 67% precision; the more conservative $\tau_z = 1.5$ default trades precision for fewer false positives. The remaining false negatives concentrate on the four-date warm-up window where no streaming statistic can discriminate without trailing data.

τ_z	TPR (recall)	FPR	Precision
0.00 (Youden-optimal)	0.741	0.119	0.669
0.75	0.696	0.101	0.690
1.50 (conservative default)	0.670	0.096	0.694
3.00	0.545	0.078	0.693
6.00	0.312	0.035	0.745

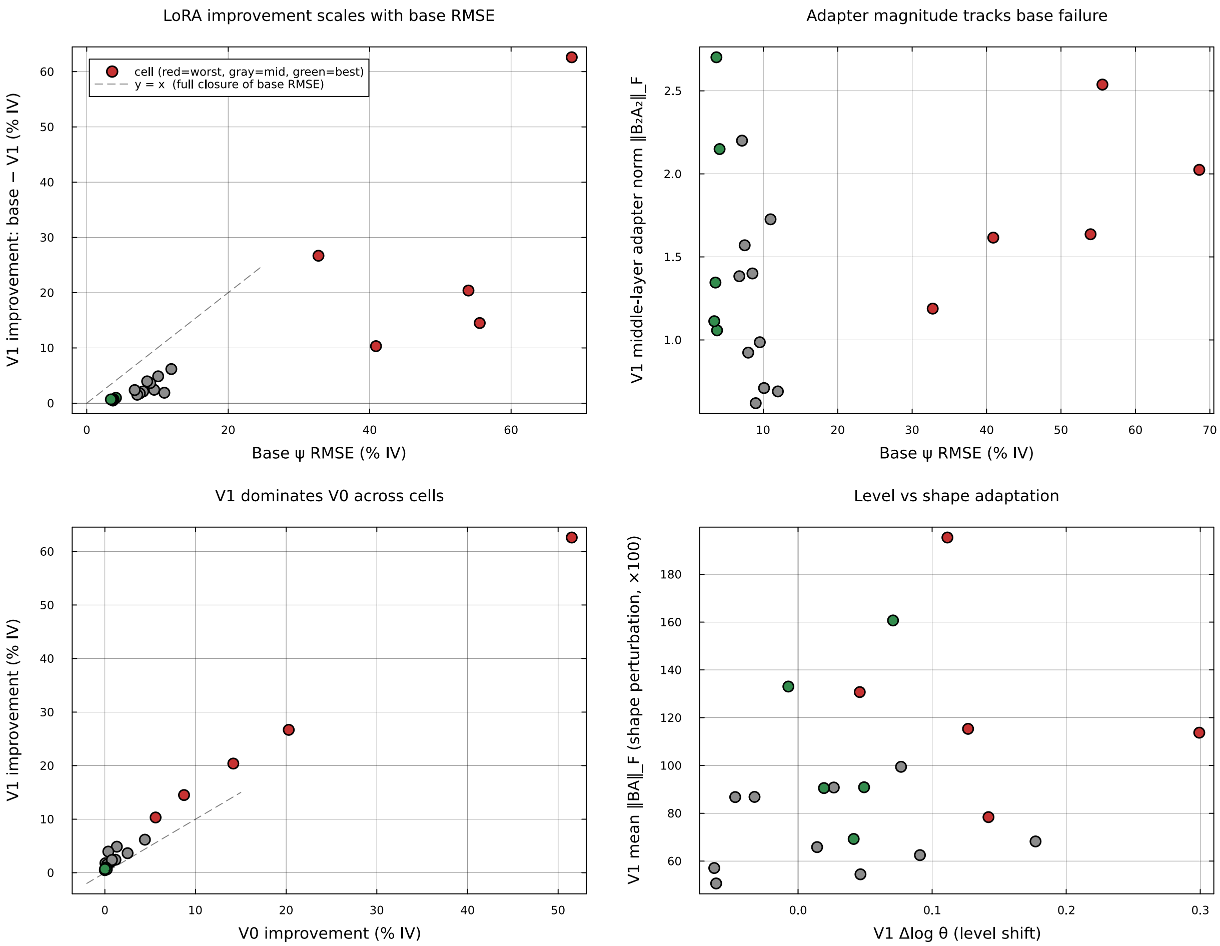

Figure S5: LoRA sweep diagnostics across 20 (ticker, date) cells. (A) V1 improvement scales with base RMSE: red points are problem cells, gray middle, green controls. The diagonal marks full-closure (V1 RMSE = 0). (B) The middle-layer adapter norm $\|B_2A_2\|_F$ tracks base RMSE, consistent with the LoRA-on-transformers pattern of intermediate-representation adaptation. (C) V1 dominates V0 across cells; the diagonal would mark equal improvement. (D) V1 partitions its budget between a small level shift ($\Delta \log \theta$ near zero on most cells) and a larger shape perturbation (mean $\|BA\|_F$), with the four shape-heavy outliers being the four highest-RMSE problem cells.

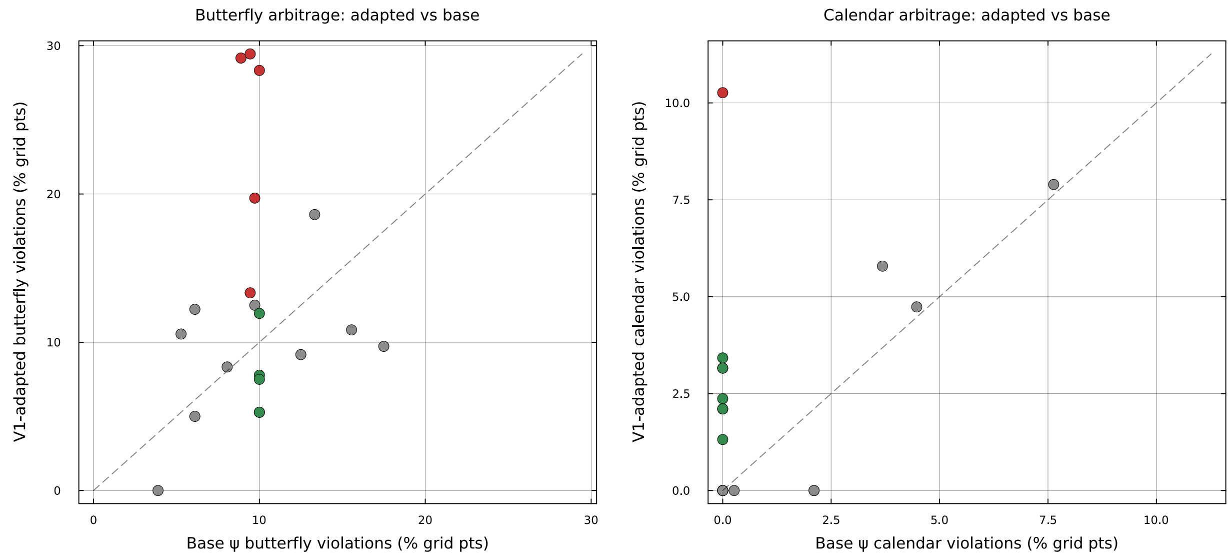

Figure S6: No-arbitrage audit on V1-adapted surfaces. Left: butterfly violation rates (% of 20×20 grid points where $\partial^2 C/\partial K^2 < 0$) for the V1-adapted surface versus the base, across the 20 sweep cells. Right: calendar violation rates (% where $\partial C/\partial T < 0$). Red = problem cells, gray = middle, green = control. The dashed line is parity; points above amplify violations, points below reduce them. Six of twenty cells lie below parity; the rest sit slightly above, with three Tech-04-23 cells producing the largest amplification.

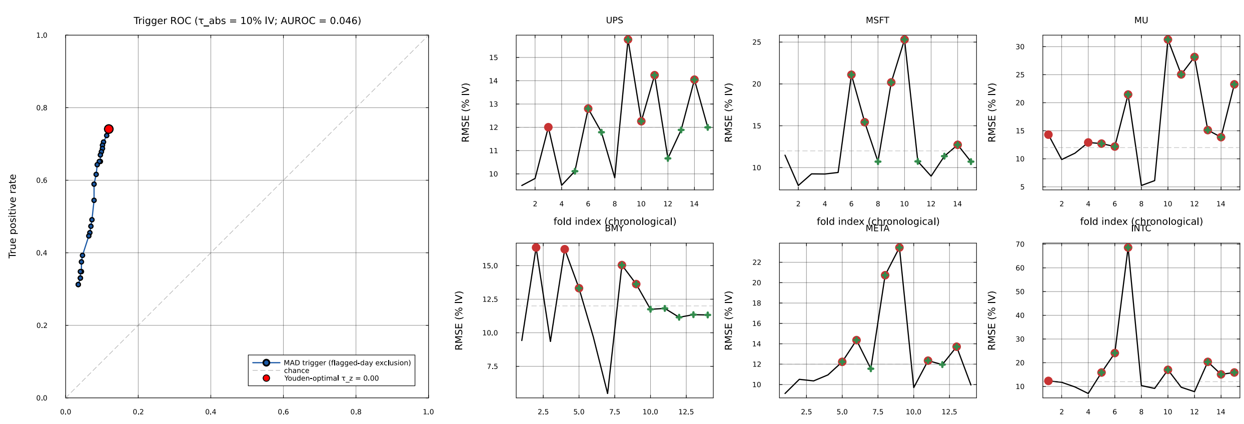

Figure S7: Streaming refit trigger. Left: ROC curve obtained by sweeping τ_z at fixed $\tau_{\text{abs}} = 10\%$ IV; the Youden-optimal operating point is marked red, and the curve is bounded away from TPR $= 1$ by the four-date warm-up window (which contains $\sim 27\%$ of truth-positive cells, none of which can be flagged without trailing data). Right: per-ticker chronological RMSE timelines for the six most-flagged tickers, with red filled circles marking truth-positive cells (base RMSE above the 12% refit-needed threshold) and green crosses marking the trigger's flag decisions at the conservative $\tau_z = 1.5$ default. The INTC trace shows the four-date sequence (04-21, 04-22, 04-23, 04-28) that the contamination-fix-equipped trigger correctly captures across the earnings-print event and the subsequent regime echo.